

# Lecture notes for April 9 to April 30

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## 1 Taylor's formula and Extremes

### 1.1 Mean Value Theorem

Let  $D \subset \mathbf{R}^n$  be an open domain, and  $f$  be a differentiable function on  $D$ . Suppose  $P, Q \in D$  satisfies  $\overline{PQ} \subset D$ . Let  $\gamma(t) = (1-t)P + tQ$  be the parameter equation for the line segment. Since  $F \circ \gamma : [0, 1] \rightarrow \mathbf{R}$  is differentiable, Lagrange Mean Value Theorem shows that

$$F(\gamma(1)) - F(\gamma(0)) = \frac{d}{dt}|_{t=\theta} (F(\gamma(t))) = \nabla F(\gamma(\theta)) \cdot \gamma'(\theta) = \nabla F(\gamma(\theta)) \cdot \overrightarrow{PQ}.$$

This implies

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$$|F(Q) - F(P)| \leq \sup_{\overline{PQ}} |\nabla F| \cdot |\overline{PQ}|;$$

- If  $D$  is connected and  $\nabla F \equiv 0$  on  $D$ , then  $F$  is constant. (Prove by yourself!)

### 1.2 Taylor's formula

Recall that for  $f(x)$  which has  $n$ -th derivative at  $x_0 \in \mathbf{R}$ , we have Taylor's formula with Peano remainder:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + o((x - x_0)^n), \quad x \rightarrow x_0. \quad (1.1)$$

If  $f$  has  $(n+1)$ -th derivative in a neighborhood of  $x_0$ , then we have Taylor's formula with Lagrange remainder:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(x_0 + \theta \Delta x)}{(n+1)!} (x - x_0)^{n+1}, \quad \theta \in (0, 1). \quad (1.2)$$

Now we consider the expansion of multivariate function  $f : D(\subset \mathbf{R}^n) \rightarrow \mathbf{R}$  at a point  $\mathbf{x}^0$ . Consider the one variable function  $\varphi(t) := f(\mathbf{x}^0 + t\mathbf{h})$  with

$t \in [-1, 1]$ . If we assume  $f$  is  $C^m$  in a neighborhood of  $\mathbf{x}^0$ , i.e. all  $m$ -th partial derivatives exist and all of them are continuous in a neighborhood of  $\mathbf{x}^0$ . By the previous theorem, this indeed implies the partial derivatives do not depend on the order of the differentiating variables.

$$\begin{aligned}\varphi'(t) &= \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\mathbf{x}^0 + t\mathbf{h})h_i; \\ \frac{d^m \varphi}{dt^m}(t) &= \sum_{i_1, \dots, i_m=1}^n \frac{\partial^m f}{\partial x_{i_1} \dots \partial x_{i_m}}(\mathbf{x}^0 + t\mathbf{h})h_{i_1} \dots h_{i_m}.\end{aligned}$$

Therefore, we get

$$f(\mathbf{x}^0 + \mathbf{h}) = \sum_{k=0}^m \frac{1}{k!} \left( \sum_{i_1, \dots, i_k=1}^n \frac{\partial^k f}{\partial x_{i_1} \dots \partial x_{i_k}}(\mathbf{x}^0)h_{i_1} \dots h_{i_k} \right) + o(|\mathbf{h}|^m), \quad |\mathbf{h}| \rightarrow 0$$

If we assume  $f$  is  $C^{m+1}$  in an open neighborhood of  $\mathbf{x}^0$ , then for  $\mathbf{h}$  sufficiently small,

$$\begin{aligned}f(\mathbf{x}^0 + \mathbf{h}) &= \sum_{k=0}^m \frac{1}{k!} \left( \sum_{i_1, \dots, i_k=1}^n \frac{\partial^k f}{\partial x_{i_1} \dots \partial x_{i_k}}(\mathbf{x}^0)h_{i_1} \dots h_{i_k} \right) \\ &\quad + \frac{1}{(m+1)!} \sum_{i_1, \dots, i_{m+1}=1}^{m+1} \frac{\partial^{m+1} f}{\partial x_{i_1} \dots \partial x_{i_{m+1}}}(\mathbf{x}^0 + \theta\mathbf{h})h_{i_1} \dots h_{i_{m+1}}, \quad \theta \in (0, 1).\end{aligned}$$

Let us put particular attention to the case  $n = 2$  and  $f$  is  $C^2$ , then

$$\begin{aligned}f(x, y) &= f(x_0, y_0) + \underline{f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0)} \\ &\quad + \underline{\frac{1}{2} (f''_{xx}(x_0, y_0)(x - x_0)^2 + 2f''_{xy}(x_0, y_0)(x - x_0)(y - y_0) + f''_{yy}(x_0, y_0)(y - y_0)^2)} \\ &\quad + o(|x - x_0|^2 + |y - y_0|^2), \quad (x, y) \rightarrow (x_0, y_0).\end{aligned}$$

**Example 1.1.** Find the Maclaurin's expansion up to second order of the function  $u(x, y) = e^x \cos y$ .

**Theorem 1.2** (Taylor expansion with Peano remainder). Assume  $f$  is defined in a neighborhood of  $\mathbf{0} \in \mathbb{R}^n$  and  $f$  is secondly differentiable at  $\mathbf{0}$ , i.e.  $\frac{\partial f}{\partial x_i}$  is differentiable at  $\mathbf{0}$  for  $i = 1, 2, \dots, n$ .

$$f(\mathbf{x}) = f(\mathbf{0}) + \nabla f(\mathbf{0}) \cdot \mathbf{x} + \frac{1}{2} \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2}(\mathbf{0})x_i^2 + \sum_{1 \leq i < j \leq n} \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0})x_i x_j + R(\mathbf{x})$$

with  $R(\mathbf{x}) = o(|\mathbf{x}|^2)$  as  $\mathbf{x} \rightarrow \mathbf{0}$ .

*Proof.* By definition, for  $i = 1, 2, \dots, n$

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) = \frac{\partial f}{\partial x_i}(\mathbf{0}) + \nabla \left( \frac{\partial f}{\partial x_i} \right)(\mathbf{0}) \cdot \mathbf{x} + R_i(\mathbf{x}) \quad (1.3)$$

with

$$\lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{R_i(\mathbf{x})}{|\mathbf{x}|} = 0, \quad i = 1, 2, \dots, n \quad (1.4)$$

Then,

$$\begin{aligned} f(\mathbf{x}) &= f(\mathbf{0}) + [f(x_1, 0, \dots, 0) - f(0, 0, \dots, 0)] + [f(x_1, x_2, \dots, 0) - f(x_1, 0, \dots, 0)] \\ &\quad + \dots + [f(x_1, x_2, \dots, x_n) - f(x_1, \dots, x_{n-1}, 0)] \\ &= f(\mathbf{0}) + \int_0^{x_1} \frac{\partial f}{\partial x_1}(t, 0, \dots, 0) dt + \int_0^{x_2} \frac{\partial f}{\partial x_2}(x_1, t, \dots, 0) dt + \dots + \int_0^{x_n} \frac{\partial f}{\partial x_n}(x_1, \dots, x_{n-1}, t) dt \\ &= f(\mathbf{0}) + \int_0^{x_1} \left( \frac{\partial f}{\partial x_1}(\mathbf{0}) + \frac{\partial^2 f}{\partial x_1^2}(\mathbf{0})t + R_1(t, 0, \dots, 0) \right) dt \\ &\quad + \int_0^{x_2} \left( \frac{\partial f}{\partial x_2}(\mathbf{0}) + \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{0})x_1 + \frac{\partial^2 f}{\partial x_2^2}(\mathbf{0})t + R_2(x_1, t, \dots, 0) \right) dt \\ &\quad + \dots \\ &\quad + \int_0^{x_n} \left( \frac{\partial f}{\partial x_n}(\mathbf{0}) + \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{0})x_1 + \dots + \frac{\partial^2 f}{\partial x_{n-1} \partial x_n}(\mathbf{0})x_{n-1} + \frac{\partial^2 f}{\partial x_n^2}(\mathbf{0})t + R_n(x_1, \dots, x_{n-1}, t) \right) dt \\ &= f(\mathbf{0}) + \nabla f(\mathbf{0}) \cdot \mathbf{x} + \frac{1}{2} \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2}(\mathbf{0})x_i^2 + \sum_{1 \leq i < j \leq n} \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0})x_i x_j + R(\mathbf{x}) \end{aligned}$$

Here the remainder

$$\begin{aligned} R(\mathbf{x}) &= \int_0^{x_1} R_1(t, 0, \dots, 0) dt + \int_0^{x_2} R_2(x_1, t, \dots, 0) dt + \dots + \int_0^{x_n} R_n(x_1, \dots, x_{n-1}, t) dt \\ &= \int_0^{x_1} \frac{R_1(t, 0, \dots, 0)}{t} t dt + \int_0^{x_2} \frac{R_2(x_1, t, \dots, 0)}{\sqrt{x_1^2 + t^2}} \sqrt{x_1^2 + t^2} dt + \dots \\ &\quad + \int_0^{x_n} \frac{R_n(x_1, \dots, x_{n-1}, t)}{\sqrt{x_1^2 + \dots + x_{n-1}^2 + t^2}} \sqrt{x_1^2 + \dots + x_{n-1}^2 + t^2} dt \end{aligned}$$

By (1.4), we know  $\forall \epsilon > 0$ , there exists  $\delta > 0$  such that for all  $\mathbf{x} \in [-\delta, \delta]^n$ ,

$$\left| \frac{R_i(\mathbf{x})}{|\mathbf{x}|} \right| < \epsilon, \quad i = 1, 2, \dots, n$$

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<sup>1</sup>Thanks to Peiyan Gu for raising this question in the interactive class.

thus

$$\begin{aligned}
|R(\mathbf{x})| &\leq \epsilon \left( \left| \int_0^{x_1} t dt \right| + \left| \int_0^{x_2} \sqrt{x_1^2 + t^2} dt \right| + \cdots + \left| \int_0^{x_n} \sqrt{x_1^2 + \cdots + x_{n-1}^2 + t^2} dt \right| \right) \\
&\leq \epsilon \left( \frac{1}{2} x_1^2 + |x_1 x_2| + \frac{1}{2} x_2^2 + \cdots + |x_n|(|x_1| + \cdots + |x_{n-1}|) + \frac{1}{2} x_n^2 \right) \\
&\leq n\epsilon |\mathbf{x}|^2
\end{aligned}$$

Therefore,  $R(\mathbf{x}) = o(|\mathbf{x}|^2)$ .  $\square$

Similarly, by Mathematical Induction, we could derive the Peano remainder for higher-order Taylor expansion.

**Corollary 1.3.** Assume  $f$  is secondly differentiable at  $\mathbf{x}_0$ , then

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{x}_0)$$

### 1.3 The extreme problem on open domain

**Definition 1.4** (Local maximum). Suppose  $f$  is defined on an open domain  $D \subset \mathbf{R}^n$ . Let  $\mathbf{x}^0 \in D$ . If  $\exists B(\mathbf{x}^0, \delta) \subset D$  such that

$$f(\mathbf{x}) \leq f(\mathbf{x}^0), \forall \mathbf{x} \in B(\mathbf{x}^0, \delta),$$

then  $\mathbf{x}^0$  is called a local maximum of  $f$ .

Similarly we can define local minimum, both local maximum and local minimum are called local extrema.

**Definition 1.5** (Critical point). If  $f$  is differentiable at  $\mathbf{x}^0$  and  $df|_{\mathbf{x}^0} = 0$ , then  $\mathbf{x}^0$  is called a critical point of  $f$ .

It is obvious that for a differentiable function, local extremal point must be critical point. The converse is in general not true. How to judge in terms of second order partial derivatives?

Let  $f \in C^2$  and  $\mathbf{x}^0$  be a critical point, then

$$f(\mathbf{x}) = f(\mathbf{x}^0) + \frac{1}{2} Q(\mathbf{h}) + R(\mathbf{h}), \quad (1.5)$$

where  $Q(\mathbf{h}) = \sum_{i,j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}^0) h_i h_j$  is a quadratic form of  $\mathbf{h}$  and  $R(\mathbf{h}) = o(|\mathbf{h}|^2)$  as  $\mathbf{h} \rightarrow \mathbf{0}$ .

We call the symmetric matrix

$$\text{Hess}(f)|_{\mathbf{x}^0} = \left( \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}^0) \right)_{i,j=1,2,\dots,n}$$

the Hessian of  $f$  at  $\mathbf{x}^0$ , then the term

$$Q(\mathbf{h}) = \mathbf{h}^T \cdot \text{Hess}(f)|_{\mathbf{x}^0} \cdot \mathbf{h}$$

is a quadratic form studied in linear algebra. In a more intrinsic way,  $\nabla^2 f|_{\mathbf{x}^0}$  is a symmetric bilinear form on  $T_{\mathbf{x}^0}\mathbf{R}^n = \mathbf{R}^n$ , and

$$\nabla^2 f|_{\mathbf{x}^0}(\mathbf{u}, \mathbf{v}) \approx \frac{f(\mathbf{x}^0 + \mathbf{u} + \mathbf{v}) - f(\mathbf{x}^0 + \mathbf{u}) - f(\mathbf{x}^0 + \mathbf{v}) + f(\mathbf{x}^0)}{\sqrt{|\mathbf{u}|^2|\mathbf{v}|^2 - (\mathbf{u} \cdot \mathbf{v})^2}}.$$

Under the standard Cartesian coordinate and basis  $\mathbf{e}_1, \dots, \mathbf{e}_n$  the quadratic form is represented by the symmetric matrix  $\text{Hess}(f)|_{\mathbf{x}^0}$ .

**Theorem 1.6.** *Let  $f$  be  $C^2$  near  $\mathbf{x}^0$  and  $\mathbf{x}^0$  be a critical point of  $f$ ,*

1. *If  $\text{Hess}(f)|_{\mathbf{x}^0}$  is positive (negative) definite, then  $\mathbf{x}^0$  is a local minimum (maximum);*
2. *If  $\text{Hess}(f)|_{\mathbf{x}^0}$  is nondegenerate but not definite, then  $\mathbf{x}^0$  is not local extremal;*
3. *If  $\text{Hess}(f)|_{\mathbf{x}^0}$  is degenerate, then we cannot tell.*

*Proof.* Suppose  $\mathbf{x}^0 = \mathbf{0}$  for simplicity. Notice that

$$f(\mathbf{x}^0 + \mathbf{h}) = f(\mathbf{x}^0) + \frac{1}{2}|\mathbf{h}|^2 \left( \sum_{i,j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}^0) \frac{h_i}{|\mathbf{h}|} \cdot \frac{h_j}{|\mathbf{h}|} + o(1) \right) := |\mathbf{h}|^2 \left( \tilde{Q}(\mathbf{h}) + o(1) \right).$$

Here,  $\tilde{Q}$  is a continuous function on  $B(\mathbf{x}^0, \delta) \setminus \{\mathbf{x}^0\}$  which is also homogeneous in the sense that  $\tilde{Q}(\lambda \mathbf{h}) = \tilde{Q}(\mathbf{h})$  for any  $\lambda > 0$  therefore  $\text{Im}(\tilde{Q}) = \text{Im}(\tilde{Q}|_{\partial B(\mathbf{x}^0, \delta)})$ . By the property of continuous function, the maximum and minimum of  $\tilde{Q}$  can be achieved on  $\partial B(\mathbf{x}^0, \delta)$ .

- If  $\text{Hess}(f)|_{\mathbf{x}^0}$  is positive definite, then  $\tilde{Q}|_{\partial B(\mathbf{x}^0, \delta)} \geq m > 0$ . Then there exists  $\delta' > 0$  such that  $|o(1)| < \frac{m}{2}$  on  $B(\mathbf{x}^0, \delta')$ . This proves that  $f(\mathbf{x}^0 + \mathbf{h}) > f(\mathbf{x}^0)$  for  $\mathbf{h} \in B(\mathbf{x}^0, \delta') \setminus \{\mathbf{x}^0\}$ ;
- If  $\text{Hess}(f)|_{\mathbf{x}^0}$  is nondegenerate but not definite, then there exists  $\mathbf{h}_1, \mathbf{h}_2$  unit vector such that  $\tilde{Q}(\mathbf{h}_1) > 0, \tilde{Q}(\mathbf{h}_2) < 0$ . Then for  $t$  sufficiently small,

$$\begin{aligned} f(\mathbf{x}^0 + t\mathbf{h}_1) &> f(\mathbf{x}^0) \\ f(\mathbf{x}^0 + t\mathbf{h}_2) &< f(\mathbf{x}^0), \end{aligned}$$

which verifies that  $\mathbf{x}^0$  is not local extremal of  $f$ .

- For the degenerate case, just notice the examples  $f_1(x, y) = x^2 + y^4, f_2(x, y) = x^2 - y^4, f_3(x, y) = x^2 + y^3$ .

□

The geometric meaning for the positive definiteness of  $\text{Hess}(f)|_{\mathbf{x}^0}$  is that  $f$  restricted to each line  $\gamma_{\mathbf{v}}(t) = t\mathbf{v}$  passing through  $\mathbf{x}^0$  at  $t = 0$  is convex at  $\mathbf{x}^0$ . We can choose another Cartesian coordinate  $\{\tilde{x}_1, \dots, \tilde{x}_n\}$  such that  $\text{Hess}(f)|_{\mathbf{x}^0}$  is diagonalized as  $\{\lambda_1, \dots, \lambda_n\}$  under this new basis. The function is of the form

$$f(\mathbf{x}^0 + \mathbf{h}) = f(\mathbf{x}^0) + df|_{\mathbf{x}^0}(\mathbf{h}) + (\lambda_1 h_1^2 + \dots + \lambda_n h_n^2) + o(|\mathbf{h}|^2), \quad \mathbf{h} \rightarrow \mathbf{0}$$

under this new coordinate.

**Example 1.7.** Find all the critical points of  $f(x, y) = x^3 - y^3 + 3x^2 + 3y^2 - 9x$  and all of its local extremals.

**Solution.** There are four critical points  $M_1 = (1, 0), M_2 = (1, 2), M_3 = (-3, 0), M_4 = (-3, 2)$ .

$M_1$  is local minimum,  $M_2, M_3$  not local extremal,  $M_4$  is local maximum.

## 1.4 Constraint extremal problem

One typical problem is to find the (local) extremal of the function  $F(x_1, \dots, x_n)$  under the *constraint condition* " $G(\mathbf{x}_1, \dots, \mathbf{x}_n) = 0$ ".

Suppose  $\nabla G(p) \neq 0$ , then  $\{G = 0\}$  can be represented by the graph of a  $n - 1$ -variables function locally around  $p$ . Saying  $G'_{x_n}(p) \neq 0$ , then there exists an open neighborhood  $U$  of  $p$  such that

$$\{G = 0\} \cap U = \{(x_1, \dots, x_{n-1}, \varphi(x_1, \dots, x_{n-1})) | (x_1, \dots, x_{n-1}) \in V\}$$

for some open set  $V \subset \mathbf{R}^{n-1}$ . This means that locally  $F(\mathbf{x})$  subject to the constraint condition  $G(\mathbf{x}) = 0$  can be understood as the function

$$\tilde{F}(x_1, \dots, x_{n-1}) := F(x_1, \dots, x_{n-1}, \varphi(x_1, \dots, x_{n-1})),$$

depending on  $(n - 1)$ -free variables  $(x_1, \dots, x_{n-1}) \in V$ . The point  $p \in \{G = 0\} \cap U$  is a constraint critical point of  $F$  iff  $(x_1, \dots, x_{n-1})$  satisfies the critical point equation:

$$\tilde{F}'_{x_i} = F'_{x_i} + F'_{x_n} \varphi'_{x_i} = 0, \quad i = 1, 2, \dots, n - 1,$$

which can be written as

$$F'_{x_i} - \frac{G'_{x_i}}{G'_{x_n}} F'_{x_n} = 0, \quad i = 1, 2, \dots, n - 1$$

according to the differentiation rule of implicit function/ implicit differentiation. The condition implies

$$\left( F'_{x_1}, \dots, F'_{x_{n-1}}, F'_{x_n} \right) (P) = \frac{F'_{x_n}}{G'_{x_n}} (P) \left( G'_{x_1}, \dots, G'_{x_{n-1}}, G'_{x_n} \right) (P).$$

Similarly, if  $F'_{x_j}(P) \neq 0$  and  $P$  is a constraint critical point, we can also derive that

$$\left(F'_{x_1}, \dots, F'_{x_{n-1}}, F'_{x_n}\right)(P) = \frac{F'_{x_j}}{G'_{x_j}}(P) \left(G'_{x_1}, \dots, G'_{x_{n-1}}, G'_{x_n}\right)(P).$$

In conclusion, if  $P$  is a constraint critical point then there exists  $\lambda \in \mathbf{R}$  such that

$$\begin{aligned} \nabla F(P) + \lambda G(P) &= 0 \\ G(P) &= 0. \end{aligned} \tag{1.6}$$

More concisely, this can be written as

$$\nabla L(P, \lambda) = 0, \tag{1.7}$$

where  $L(P, \lambda) := F(P) + \lambda G(P)$  is a function encoding the constraint equation to the original function in question via introducing one more variable. This method of looking for conditional critical points is called “Lagrange Multiplier Method”.

1. This method can find all conditional critical points of  $F$ ;
2. The solution to (1.7) might contain some “singular point”  $P$  on the constraint set, i.e.  $G(P) = 0, \nabla G(P) = 0$ . This type of solution satisfies  $\nabla F(P) = 0$ , therefore it is critical without constraint condition.
3. Geometrically, the level set of  $F$  must be tangential to the level set of the constraint function  $G$  at any critical point, i.e.

$$\ker dG|_P \subset \ker dF|_P.$$

At least at any smooth point of the constraint set  $\{G = 0\}$ , this condition is an equivalent condition for the “critical point”.

**Example 1.8.** Find the constraint local extremels and maximum and minimum for the function  $z = xy$  under the condition  $(x - 1)^2 + y^2 = 1$ .

**Solution.** Consider the Lagrange function  $L(x, y, \lambda) := xy + \lambda((x - 1)^2 + y^2 - 1)$ . The Lagrange Multiplier Method gives the equation

$$\begin{aligned} y + 2\lambda(x - 1) &= 0 \\ x + 2\lambda y &= 0 \\ (x - 1)^2 + y^2 - 1 &= 0. \end{aligned}$$

Plugging the second equation to the first and the third, we can get that  $x = -2\lambda y$  and  $(1 + 4\lambda^2)y^2 = -4\lambda y$ . If  $y = 0$ , then  $x = 0$  and  $\lambda = 0$ ; if  $y \neq 0$ , we get  $\lambda = \pm \frac{\sqrt{3}}{2}$  and  $\lambda = \mp \frac{\sqrt{3}}{2}$  and  $x = \frac{3}{2}$ .

The function  $z$  has 3 conditional critical points. Since  $z|_{(x,y)=(0,0)} = 0$ ,  $z|_{(x,y)=(\frac{3}{2}, \frac{\sqrt{3}}{2})} = \frac{3\sqrt{3}}{4}$  and  $z|_{(\frac{3}{2}, -\frac{\sqrt{3}}{2})} = -\frac{3\sqrt{3}}{4}$ . Moreover,  $z = xy$  can take both positive and negative values near  $(0,0)$ . We conclude that  $\frac{3}{2}, \frac{\sqrt{3}}{2}$  is the maximal point and  $\frac{3}{2}, -\frac{\sqrt{3}}{2}$  is the minimal point, and  $(0,0)$  is not local extremal.

**Example 1.9.** Find the distance from  $(0,0,0)$  to the surface defined by  $(x-y)^2 - z^2 = 1$ .

**Solution.** It amounts to find the minimum of the function

$$F(x, y, z) = \rho((x, y, z), (0, 0, 0))^2 = x^2 + y^2 + z^2$$

under the constraint condition  $(x-y)^2 - z^2 = 1$ .

Consider the Lagrange Multiplier Method: define  $L(x, y, z, \lambda) = (x^2 + y^2 + z^2) + \lambda((x-y)^2 - z^2 - 1)$  and solve the equations

$$\begin{aligned} 2x + 2\lambda(x-y) &= 0 \\ 2y - 2\lambda(x-y) &= 0 \\ 2z - 2\lambda z &= 0 \\ (x-y)^2 - x^2 - 1 &= 0. \end{aligned}$$

We obtain that  $\lambda = -\frac{1}{2}$ , and  $(x, y, z) = (\frac{1}{2}, -\frac{1}{2}, 0)$  or  $(-\frac{1}{2}, \frac{1}{2}, 0)$ . Since these two points are the only two critical points and  $F$  takes the same value  $\frac{1}{2}$  on them and meanwhile  $F$  is unbounded from above on the constraint surface. We know that the two constraint critical points are minimal points.

Geometrically we know that  $(x-y)^2 - z^2 = 1$  is a hyperbolic cylinder perpendicular to the  $Oxy$  plane, and it is obvious that the distance from the origin to this surface is realized by the two vertices of the hyperbola in the  $Oxy$  plane.

**Example 1.10.** Given a plane  $\Pi : \ell(x, y, z) = ax + by + cz + d = 0$  and  $n$  points  $P_i = (x_i, y_i, z_i)$  for  $i = 1, 2, \dots, n$ . Find the point(s) on  $\Pi$  which realizes the minimum of the summed square distance.

**Solution.** Define the function

$$u(x, y, z) = \sum_{i=1}^n |PP_i|^2 = \sum_{i=1}^n [(x-x_i)^2 + (y-y_i)^2 + (z-z_i)^2],$$

we would like to seek for the constraint minimum of  $u$  under the condition  $\ell = 0$ .

The Lagrange Multiplier Method: Consider the function  $L(x, y, z, \lambda) = u(x, y, z) +$



$\lambda \ell(x, y, z)$ . The critical point equation of  $L$  gives the equations

$$\begin{aligned} L'_x &= \sum_{i=1}^n 2(x - x_i) + \lambda a = 0 \\ L'_y &= \sum_{i=1}^n 2(y - y_i) + \lambda b = 0 \\ L'_z &= \sum_{i=1}^n 2(z - z_i) + \lambda c = 0 \\ L'_\lambda &= ax + by + cz + d = 0 \end{aligned}$$

This gives the solution

$$\begin{aligned} \lambda &= \frac{\ell(\bar{x}, \bar{y}, \bar{z})}{|\mathbf{n}|^2} \cdot 2n \\ (x, y, z) &= (\bar{x}, \bar{y}, \bar{z}) - \frac{\ell(\bar{x}, \bar{y}, \bar{z})}{|\mathbf{n}|^2} \mathbf{n}. \end{aligned}$$

where  $(\bar{x}, \bar{y}, \bar{z}) = \frac{1}{n} (\sum_{i=1}^n x_i, \sum_{i=1}^n y_i, \sum_{i=1}^n z_i)$  is the center of gravity of  $P_1, \dots, P_n$ . This is precisely the projection of the center of gravity on the plane.

This result can also be achieved by direct algebraic method by completing squares. By looking at the projection points  $\hat{P}_i$  on  $\Pi$ , we can reduce to the case when all points lie on one plane.

**Example 1.11.** Find the extreme of the function  $u(x_1, \dots, x_n) = x_1 \cdots x_n$  under the constraint equation  $x_1 + \cdots + x_n = a$  ( $a > 0$ ).

**Solution.** Consider the Lagrange Multiplier Method:  $L(x_1, \dots, x_n) = x_1 \cdots x_n + \lambda(x_1 + \cdots + x_n - a)$ .

$$\begin{aligned} L'_{x_1} &= x_2 \cdots x_n + \lambda = 0 \\ &\dots\dots\dots \\ L'_{x_n} &= x_1 \cdots x_{n-1} + \lambda = 0 \\ L'_\lambda &= x_1 + \cdots + x_n - a = 0. \end{aligned}$$

We have two type of solutions. The first type is  $\lambda = 0$  and at least two of the variables  $x_i$ 's are 0, while the others satisfies  $x_1 + \cdots + \hat{x}_i + \cdots + \hat{x}_j + \cdots + x_n = a$ . Near these points it is easily seen that the sign of  $u$  can change, therefore those points are not local extreme. The second type of solution is  $\lambda \neq 0$ , this implies that  $x_i = \frac{a}{n}$  for  $i = 1, 2, \dots, n$ . As  $u$  is positive at this point and  $u$  attains 0 on the boundary of the quadrant  $x_1 + \cdots + x_n = a, x_1, \dots, x_n \geq 0$ , therefore the point  $(\frac{a}{n}, \dots, \frac{a}{n})$  is the maximum of this set.

It is easily seen that for  $n \geq 3$ , the function has no maximum on the whole constraint set  $x_1 + \cdots + x_n = a$ .

## 1.5 Multi-constraints problem

First, let us consider the function  $F(x, y, z)$  under the constraint  $G(x, y, z) = H(x, y, z) = 0$ . More generally, we can consider the extremal value problem for the function  $F(\mathbf{x})$  under the constraint equation  $\mathbf{G}(\mathbf{x}) = \mathbf{0}$ . Using the geometrical interpretation,  $P$  is a critical point under the constraint iff

$$\ker d\mathbf{G}|_{\mathbf{x}} \subset \ker dF|_{\mathbf{x}}. \quad (1.8)$$

This is because of the following : the point  $P \in \{G = 0\}$  is a constraint critical point of  $F$  iff for any smooth curve  $\gamma \subset \{G = 0\}$  passing through a point  $P$  (at  $t = 0$ ) we have  $F \circ \gamma$  has  $t = 0$  as a critical point, i.e.  $dF|_P(\gamma'(0)) = 0$ .

The condition (1.8) is equivalent to

$$(\ker dF|_{\mathbf{x}})^\perp \subset (\ker d\mathbf{G}|_{\mathbf{x}})^\perp, \quad (1.9)$$

which is further equivalent to

$$\nabla F(\mathbf{x}) \in \text{Span}\{\nabla G_1(\mathbf{x}), \dots, \nabla G_m(\mathbf{x})\}. \quad (1.10)$$

This leads to the **Lagrange Multiplier Method**: define  $L(\mathbf{x}, \boldsymbol{\lambda}) = F(\mathbf{x}) + \boldsymbol{\lambda} \cdot \mathbf{G}(\mathbf{x})$ , then the constraint critical point problem for  $F$  under the constraint  $\mathbf{G}$  is equivalent to the “unconditioned critical point problem”:

$$\nabla L(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{0}.$$

The vector  $\boldsymbol{\lambda}$  is called the Lagrange Multipliers.

**Question 1.12.** *Via the use of Implicit Function Theorem, verify the Lagrange Multiplier Method for multi-constraint condition.*

**Question 1.13.** *How to judge iff a constraint critical point is local extrema?*

$$\begin{aligned} \frac{d}{dx} F(x, \varphi(x)) &= \frac{\frac{\partial(F, G)}{\partial(x, y)}}{\frac{\partial G}{\partial y}} \\ \frac{d^2}{dx^2} F(x, \varphi(x)) &= \frac{\frac{\partial(\frac{\partial(F, G)}{\partial(x, y)}, G)}{\partial(x, y)}}{\left(\frac{\partial G}{\partial y}\right)^3} \end{aligned}$$

We can compute the sign of the second derivative. Try to find a geometric interpretation of this “constraint second derivative”!

**Theorem 1.14 (Constraint Hessian Criteria).** *For a constraint critical point  $\mathbf{x}$ , we can look at the Hessian of  $L$  at  $(\mathbf{x}, \boldsymbol{\lambda})$  restricted to the subspace  $\ker d\mathbf{G}|_{\mathbf{x}}$ .*

If this restricted quadratic form is positive definite, i.e.  $\text{Hess}(F) + \lambda \text{Hess}(G)$  is positive definite on  $\ker dG|_{\mathbf{x}}$ , then  $\mathbf{x}$  is constraint local minimum.

For multi-constraint, if  $\text{Hess}(F) + \lambda \text{Hess}(G)$  is positive definite on  $\ker d\mathbf{G}|_{\mathbf{x}}$ , then  $\mathbf{x}$  is a constraint local minimum.

*Proof.* • Suppose at this constraint critical point  $G'_n(\mathbf{x}) \neq 0$ , then we can compute the Hessian of the composite  $(n-1)$ -variables function  $\tilde{F}$ . For  $1 \leq i, j \leq n-1$ ,

$$\begin{aligned} \frac{\partial^2 \tilde{F}}{\partial x_i \partial x_j} &= \left( F_{ij} - \frac{F_n}{G_n} G_{ij} \right) - \left( F_{in} - \frac{F_n}{G_n} G_{in} \right) \frac{G_j}{G_n} \\ &\quad - \left( F_{nj} - \frac{F_n}{G_n} G_{nj} \right) \frac{G_i}{G_n} + \left( F_{nn} - \frac{F_n}{G_n} G_{nn} \right) \frac{G_i G_j}{G_n^2} \end{aligned}$$

On the other hand, if we denote  $\boldsymbol{\xi} = (\xi^1, \dots, \xi^n)^T = (\hat{\boldsymbol{\xi}}, \xi^n)^T$  and  $\nabla G(\mathbf{x}) = (\hat{\nabla} G(\mathbf{x}), G_n(\mathbf{x}))$ , then  $\boldsymbol{\xi} \in \ker dG|_{\mathbf{x}}$  iff

$$\xi^n = -\frac{\hat{\boldsymbol{\xi}} \cdot \hat{\nabla} G}{G_n}.$$

Because at the constraint critical point  $\mathbf{x}$ , it holds that

$$\nabla F(\mathbf{x}) + \lambda \nabla G(\mathbf{x}) = 0,$$

we have

$$\begin{aligned} &\boldsymbol{\xi}^T \left( \frac{\partial^2 F}{\partial x_\alpha \partial x_\beta} + \lambda \frac{\partial^2 G}{\partial x_\alpha \partial x_\beta} \right) \boldsymbol{\xi} \\ &= \xi^\alpha (F_{\alpha\beta} + \lambda G_{\alpha\beta}) \xi^\beta = (F_{ij} + \lambda G_{ij}) \xi^i \xi^j - (F_{in} + \lambda G_{in}) \xi^i \frac{\hat{\boldsymbol{\xi}} \cdot \hat{\nabla} G}{G_n} \\ &\quad - (F_{nj} + \lambda G_{nj}) \xi^j \frac{\hat{\boldsymbol{\xi}} \cdot \hat{\nabla} G}{G_n} + (F_{nn} + \lambda G_{nn}) \frac{(\hat{\boldsymbol{\xi}} \cdot \hat{\nabla} G)^2}{G_n^2} \\ &= \boldsymbol{\xi}^T \left( \frac{\partial^2 \tilde{F}}{\partial x_i \partial x_j} \right) \hat{\boldsymbol{\xi}} \end{aligned}$$

- For a more concise proof. Let  $\gamma : (-\epsilon, \epsilon) \rightarrow \mathbf{R}^n$  be any smooth curve lying in the constraint set  $\mathcal{S} = \{\mathbf{G} = 0\}$  and passing through the critical point  $\mathbf{x}$ . Then the formulae

$$\begin{aligned} \frac{d}{dt} F(\gamma(t)) &= \sum_{i=1}^n \frac{\partial F}{\partial x_i}(\gamma(t)) \gamma'_i(t), \\ \frac{d^2}{dt^2} F(\gamma(t)) &= \frac{\partial^2 F}{\partial x_i \partial x_j}(\gamma(t)) \gamma'_i(t) \gamma'_j(t) + \frac{\partial F}{\partial x_i}(\gamma(t)) \gamma''_i(t) \end{aligned}$$

implies that

$$\sum_{i=1}^n \frac{\partial F}{\partial x_i}(\mathbf{x}) \gamma'_i(0) = 0, \quad \forall \gamma'(0) \in T_{\mathbf{x}}\mathcal{S} = \ker d\mathbf{G}|_{\mathbf{x}} = \cap_{k=1}^m (\nabla G_k|_{\mathbf{x}})^\perp.$$

This implies there exists  $\lambda_1, \dots, \lambda_m$  such that

$$\nabla F|_{\mathbf{x}} = \lambda_1 \nabla G_1|_{\mathbf{x}} + \dots + \lambda_m \nabla G_m|_{\mathbf{x}}.$$

We also have

$$\begin{aligned} \frac{d^2}{dt^2} \Big|_{t=0} F(\gamma(t)) &= \sum_{i,j=1}^n \frac{\partial^2}{\partial x_i \partial x_j}(\mathbf{x}) \gamma'_i(0) \gamma'_j(0) + \sum_{i=1}^n \sum_{k=1}^m \lambda_k \frac{\partial G_k}{\partial x_i}(\mathbf{x}) \gamma''_i(0) \\ &= \sum_{i,j=1}^n \frac{\partial^2 F}{\partial x_i \partial x_j}(\mathbf{x}) \gamma'_i(0) \gamma'_j(0) - \sum_{k=1}^m \lambda_k \sum_{i,j=1}^n \frac{\partial^2 G_k}{\partial x_i \partial x_j}(\mathbf{x}) \gamma'_i(0) \gamma'_j(0) \\ &= \left( \nabla^2 F - \sum_{k=1}^m \lambda_k \nabla^2 G_k \right) (\gamma'(0), \gamma'(0)). \end{aligned}$$

If  $\nabla^2 F - \sum_{k=1}^m \lambda_k \nabla^2 G_k$  is positive definite on  $T_{\mathbf{x}}\mathcal{S}$ , then

$$\frac{d^2}{dt^2} \Big|_{t=0} F(\gamma(t)) > 0.$$

The conclusion follows.  $\square$

**Problem 1.** Let  $\mathbf{G} : M_{3 \times 3}(\mathbf{R}) = \mathbf{R}^9 \rightarrow \text{Sym}_{3 \times 3}(\mathbf{R}) = \mathbf{R}^6$  be the map  $A \mapsto AA^T - I$ . Let  $F : M_{3 \times 3}(\mathbf{R}) \rightarrow \mathbf{R}$  be the function  $A \mapsto \text{Tr } A$ . Find all critical point of  $F$  under the constraint condition  $\mathbf{G}(A) = \mathbf{0}$ .

## 1.6 Coordinate covariance

Let  $\mathbf{x} = \mathbf{x}(\mathbf{y})$ , i.e.  $x_i = x_i(y_1, \dots, y_n)$  for  $i = 1, \dots, n$  be a coordinate transformation, then we have

$$\begin{aligned} \frac{\partial f}{\partial y_i} &= \frac{\partial f}{\partial x_k} \frac{\partial x_k}{\partial y_i} \\ \frac{\partial^2 f}{\partial y_i \partial y_j} &= \frac{\partial}{\partial y_j} \left( \frac{\partial f}{\partial x_k} \frac{\partial x_k}{\partial y_i} \right) \\ &= \frac{\partial^2 f}{\partial x_k \partial x_l} \frac{\partial x_k}{\partial y_i} \frac{\partial x_l}{\partial y_j} + \frac{\partial f}{\partial x_k} \frac{\partial^2 x_k}{\partial y_i \partial y_j}. \end{aligned}$$

The equation shows that

$$\left( \frac{\partial f}{\partial y_1}, \dots, \frac{\partial f}{\partial y_n} \right) = \left( \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \dots & \frac{\partial x_1}{\partial y_n} \\ \dots & \dots & \dots \\ \frac{\partial x_n}{\partial y_1} & \dots & \frac{\partial x_n}{\partial y_n} \end{pmatrix} \quad (1.11)$$

but

$$\left( \frac{\partial^2 f}{\partial y_i \partial y_j} \right) (\mathbf{y}) = \left( J_{\mathbf{y}}(\mathbf{x}) \right)^T \left( \frac{\partial^2 f}{\partial x_i \partial x_j} \right) |_{\mathbf{x}=\mathbf{x}(\mathbf{y})} J_{\mathbf{y}}(\mathbf{x})$$

does not hold in general. In this sense, we say that first partial derivatives do not depend on particular coordinate system while the second partial derivatives depend on particular coordinate system. A historical way of this difficulty is the “introducing” of correction terms by first derivatives, making the “new second derivatives” transform covariantly. More precisely speaking, consider

$$f_{,ij} := f_{ij} - \Gamma_{ij}^k f_k,$$

where the *symbols*  $\Gamma_{ij}^k$  are suitable functions.

## 2 Curves and Surfaces in $\mathbf{R}^3$

### 2.1 Parametric curve

The parametric equation of a line in  $\mathbf{R}^3$  is given by:  $\mathbf{r}(t) = P_0 + t\mathbf{v}$  for  $t \in \mathbf{R}$ .

In general, we call a vector-valued function  $\mathbf{r} : [\alpha, \beta] \rightarrow \mathbf{R}^3$  given by  $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$  a parametric curve in  $\mathbf{R}^3$ . If  $\mathbf{r}$  is  $C^1$ , i.e.  $x'(t), y'(t), z'(t)$  are continuously differentiable.

For a  $C^1$  curve, define  $\mathbf{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t}$  is called the tangent vector of the parametric curve at  $\mathbf{r}(t)$ . In terms of differential,  $d\mathbf{r} = \mathbf{r}'(t)dt = (x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k})dt$ . If  $\mathbf{r}'(t_0) \neq 0$ , then  $\mathbf{r}(t)$  is called a regular point of the curve. If  $\mathbf{r}'(t_0) = 0$ , we say  $\mathbf{r}(t_0)$  is a singular point. A regular curve is a curve for which every point is regular. The normal plane to a curve at  $\mathbf{r}(t_0)$  is defined as  $(\mathbf{r} - \mathbf{r}(t_0)) \cdot \mathbf{r}'(t_0) = 0$ .

We’ve discussed the arc length of a curve: the idea is to use the length of broken line segments to approximate the curve.

For a  $C^1$  parametric curve  $\mathbf{r} : [\alpha, \beta] \rightarrow \mathbf{R}^3$ , then

$$\lim_{\|T\| \rightarrow 0} \ell(T) = \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n |M_{i-1}M_i| = \int_{\alpha}^{\beta} |\mathbf{r}'(t)| dt$$

exists, and the limit is called the *arc length* of the curve.

If  $t : [\alpha', \beta'] \rightarrow [\alpha, \beta]$  is a coordinate transformation of the domain, then  $\tilde{\mathbf{r}} = \mathbf{r} \circ t : [\alpha', \beta'] \rightarrow [\alpha, \beta]$  is another parameterization of the same curve. We have

$$\tilde{\mathbf{r}}'(u) = \mathbf{r}'(t) \frac{dt}{du}. \quad (2.1)$$

This means the tangent vector depends on particular parametrization of the curve, but the tangent line does not depend on the particular parametrization. Using  $u$ , the arc length can be calculated as

$$\int_{\alpha'}^{\beta'} |\tilde{\mathbf{r}}'(u)| du \quad (2.2)$$

which is easily seen to be equal to  $\int_{\alpha}^{\beta} |\mathbf{r}'(t)| dt$ . Therefore, the total arc length is a geometric quantity, which is intrinsic to the curve.

Define

$$s = \int_{\alpha}^t |\mathbf{r}'(u)| du,$$

then  $s : [\alpha, \beta] \rightarrow [0, l]$  is a coordinate transformation as  $\frac{ds}{dt} = |\mathbf{r}'(t)| \neq 0$  for all  $t \in [\alpha, \beta]$ . This parameter is special as

$$\left| \frac{d\mathbf{r}}{ds} \right| = \left| \frac{d\mathbf{r}}{dt} \frac{dt}{ds} \right| = |\mathbf{r}'(t)| \cdot \left| \frac{1}{\frac{ds}{dt}} \right| = 1,$$

this parameter is called the *arc length parameter*. Under the arc length parameter, we have

$$\left( \frac{dx}{ds} \right)^2 + \left( \frac{dy}{ds} \right)^2 + \left( \frac{dz}{ds} \right)^2 = 1,$$

or formally

$$ds^2 = dx^2 + dy^2 + dz^2, \quad (2.3)$$

which measures what is the length of a small portion of the curve described by a small increase  $dx, dy, dz$  in each coordinate component.

### 2.1.1 Curvature of space curve

The curvature characterize how does the direction of a curve vary.

Take the derivative with respect to  $s$  to the equation

$$\mathbf{r}'(s) \cdot \mathbf{r}'(s) = 1,$$

we obtain that

$$\mathbf{r}''(s) \cdot \mathbf{r}'(s) = 0. \quad (2.4)$$

The vector  $\mathbf{r}''(s) \in N_{\mathbf{r}(s)}(\gamma)$  is called the *principal normal vector*. If  $\mathbf{r}''(s) \neq 0$ , then  $\boldsymbol{\kappa}(s) := \mathbf{r}'(s) \times \mathbf{r}''(s) \in N_{\mathbf{r}(s)}(\gamma)$  gives another normal vector to  $\gamma$ , this is called the *secondary normal*.

Take a small increment  $\Delta s$  and let  $\Delta\alpha$  be the angle of  $\mathbf{r}'(s + \Delta s)$  and  $\mathbf{r}'(s)$ , then

$$|\mathbf{r}'(s) \times \mathbf{r}'(s + \Delta s)| = \sin \Delta\alpha.$$

As instantaneous rate of change of  $\alpha$  with respect to  $s$ , we have

$$\begin{aligned} \lim_{\Delta s \rightarrow 0} \frac{\Delta\alpha}{\Delta s} &= \lim_{\Delta s \rightarrow 0} \frac{\sin \Delta\alpha}{\Delta s} \\ &= \lim_{\Delta s \rightarrow 0} \left| \mathbf{r}'(s) \times \left( \frac{\mathbf{r}'(s + \Delta s) - \mathbf{r}'(s)}{\Delta s} \right) \right| \\ &= |\mathbf{r}'(s) \times \mathbf{r}''(s)| = |\boldsymbol{\kappa}(s)|. \end{aligned}$$

By this reason,  $\boldsymbol{\kappa}$  is called the *curvature vector* and  $\kappa(s) = |\boldsymbol{\kappa}(s)|$  is called the curvature. It measures on the one hand how fast does the direction of the curve change.

**Remark 2.1.** For a regular curve, what is the speciality of the principal normal?  
Consider the Taylor expansion:

$$\mathbf{r}(s) = \mathbf{r}(s_0) + \mathbf{r}'(s_0)(s - s_0) + \frac{1}{2}\mathbf{r}''(s_0)(s - s_0)^2 + o(|s - s_0|^2).$$

Define the local curve  $\boldsymbol{\varphi}(s) = \mathbf{r}(s_0) + \mathbf{r}'(s_0)(s - s_0) + \frac{1}{2}\mathbf{r}''(s_0)(s - s_0)^2$ , then the image of  $\boldsymbol{\varphi}$  totally lies in  $\text{Span}\{\mathbf{r}'(s_0), \mathbf{r}''(s_0)\}$ . Since  $\mathbf{r}(s) - \boldsymbol{\varphi}(s) = o((s - s_0)^2)$  as  $s \rightarrow s_0$ , the curve  $\gamma$  is  $2^{nd}$ -order tangential to the parabola  $\boldsymbol{\varphi}(s)$ . Actually, this parabola is the unique parabola in  $\mathbf{R}^3$  which is  $2^{nd}$ -order tangential to  $\gamma$  at  $\mathbf{r}(s_0)$ .

**Question 2.2.** Try to work out the formula for curvature under general parameter:

$$\kappa = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}.$$

**Example 2.3.**

- Planar curve (we've discussed last semester).
- Spring curve:  $\mathbf{r}(t) = (R \cos t, R \sin t, kt)$ . (How to distinguish this curve with a planar circle, for this curve the secondary normal is also varying while for planar curve the secondary normal direction does not vary.)

## 2.2 Parametric surface

We call a vector-valued function  $\mathbf{r} : D(\subset \mathbf{R}^2) \rightarrow \mathbf{R}^3$  a *parametric surface*. If  $\mathbf{r} \in C^1$ , then we say this parametric surface is a  $C^1$  surface. We have

$$d\mathbf{r} = \frac{\partial \mathbf{r}}{\partial u} du + \frac{\partial \mathbf{r}}{\partial v} dv := \mathbf{r}'_u du + \mathbf{r}'_v dv.$$

Here  $\mathbf{r}'_u = \frac{\partial \mathbf{r}}{\partial u}$  and  $\mathbf{r}'_v = \frac{\partial \mathbf{r}}{\partial v}$  represent the tangent vector to the  $u$ -curve and  $v$ -curve on the surface.

If  $\mathbf{r}'_u(u_0, v_0) \times \mathbf{r}'_v(u_0, v_0) \neq 0$ , the point  $(u_0, v_0)$  is called a regular point of the parametric surface.

Consider any  $C^1$  curve on  $S$  passing through  $M_0 = \mathbf{r}(u_0, v_0)$  of the form  $\mathbf{r}(u(t), v(t))$  with  $(u(t_0), v(t_0)) = (u_0, v_0)$ . The tangent vector to the curve at  $M_0$  is

$$\frac{d}{dt}\bigg|_{t=t_0} \mathbf{r}(u(t), v(t)) = \mathbf{r}'_u(u_0, v_0)u'(t_0) + \mathbf{r}'_v(u_0, v_0)v'(t_0) \in \text{Span}\{\mathbf{r}'_u(u_0, v_0), \mathbf{r}'_v(u_0, v_0)\}.$$

Conversely, at a regular point any vector in  $\text{Span}\{\mathbf{r}'_u(u_0, v_0), \mathbf{r}'_v(u_0, v_0)\}$  can be realized as the tangent vector of some curve on  $S$  passing through  $M_0$ . Therefore,  $\text{Span}\{\mathbf{r}'_u(u_0, v_0), \mathbf{r}'_v(u_0, v_0)\}$  is the *tangent plane* of  $S$  at  $M_0$ .

One normal vector of the surface at  $(u_0, v_0)$  is

$$\mathbf{r}'_u \times \mathbf{r}'_v = \frac{\partial(y, z)}{\partial(u, v)} \mathbf{i} + \frac{\partial(z, x)}{\partial(u, v)} \mathbf{j} + \frac{\partial(x, y)}{\partial(u, v)} \mathbf{k}.$$

The parallelogram spanned by  $\{\mathbf{r}'_u, \mathbf{r}'_v\}$  has area square equal to

$$|\mathbf{r}'_u \times \mathbf{r}'_v|^2 = |\mathbf{r}'_u|^2 |\mathbf{r}'_v|^2 - (\mathbf{r}'_u \cdot \mathbf{r}'_v)^2 := EG - F^2, \quad (2.5)$$

where

$$\begin{aligned} E &= \mathbf{r}'_u \cdot \mathbf{r}'_u \\ F &= \mathbf{r}'_u \cdot \mathbf{r}'_v \\ G &= \mathbf{r}'_v \cdot \mathbf{r}'_v. \end{aligned}$$

This means that a tiny rectangle spanned by  $\Delta u$  and  $\Delta v$  in the parameter region is mapped approximately to a parallelogram spanned by  $\mathbf{r}'_u \Delta u$  and  $\mathbf{r}'_v \Delta v$  on the surface with area  $\sqrt{EG - F^2} \Delta u \Delta v$ .

We define

$$\begin{aligned} d\mathbf{r} \odot d\mathbf{r} &= (\mathbf{r}'_u du + \mathbf{r}'_v dv) \cdot (\mathbf{r}'_u du + \mathbf{r}'_v dv) \\ &= Edu \otimes du + Fdu \otimes dv + Fdv \otimes du + Gdv \otimes dv, \end{aligned}$$

then  $d\mathbf{r} \odot d\mathbf{r}$  is a positive definite symmetric bilinear form on  $\mathbf{R}^2$ . This is called the *first fundamental form* of the parametric surface. This can be used to measure the length square of any tangent vector  $\mathbf{v} = a\mathbf{r}'_u + b\mathbf{r}'_v \in \text{Span}\{\mathbf{r}'_u, \mathbf{r}'_v\}$ :

$$\mathbf{v} \cdot \mathbf{v} = (a \ b) \begin{pmatrix} \mathbf{r}'_u \\ \mathbf{r}'_v \end{pmatrix} \cdot (\mathbf{r}'_u \ \mathbf{r}'_v) \begin{pmatrix} a \\ b \end{pmatrix} = (a \ b) \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = Ea^2 + 2Fab + Gb^2.$$

In this meaning,  $ds^2 = Edu^2 + 2Fdudv + Gdv^2$  is called *length element* of a surface. In the same spirit,  $\sqrt{EG - F^2}dudv$  is called the *area element*.

**Example 2.4.** Consider the parametric surface

$$\mathbf{r}(\theta, \varphi) = (R \sin \theta \cos \varphi, R \sin \theta \sin \varphi, R \cos \theta), \quad \theta \in [0, \pi], \varphi \in [0, 2\pi].$$

**Solution.** Firstly,

$$\begin{aligned} \mathbf{r}'_\theta &= (R \cos \theta \cos \varphi, R \cos \theta \sin \varphi, -R \sin \theta), \\ \mathbf{r}'_\varphi &= (-R \sin \theta \sin \varphi, R \sin \theta \cos \varphi, 0) \\ E &= \mathbf{r}'_\theta \cdot \mathbf{r}'_\theta = R^2 \\ F &= \mathbf{r}'_\theta \cdot \mathbf{r}'_\varphi = 0 \\ G &= \mathbf{r}'_\varphi \cdot \mathbf{r}'_\varphi = R^2 \sin^2 \theta \end{aligned}$$

The length element for  $S^2$  is therefore

$$ds^2 = R^2 (d\theta^2 + \sin^2 \theta d\varphi^2).$$



The normal vector at  $(\theta, \varphi)$  is given by

$$\mathbf{r}'_{\theta} \times \mathbf{r}'_{\varphi} = (R^2 \sin^2 \theta \cos \varphi, R^2 \sin^2 \theta \sin \varphi, R^2 \sin \theta \cos \theta),$$

and the unit normal vector is

$$\mathbf{n} = \frac{\mathbf{r}'_{\theta} \times \mathbf{r}'_{\varphi}}{\sqrt{EG - F^2}} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) = \frac{\mathbf{r}}{R}.$$

If we take  $\tilde{D} \rightarrow D$  a coordinate transformation given by  $(u, v) = (u(\theta, \varphi), v(\theta, \varphi))$ , then

$$d\mathbf{r} = \mathbf{r}'_{\theta} d\theta + \mathbf{r}'_{\varphi} d\varphi = \mathbf{r}'_u (u'_{\theta} d\theta + u'_{\varphi} d\varphi) + \mathbf{r}'_v (v'_{\theta} d\theta + v'_{\varphi} d\varphi).$$

Therefore we obtain that

$$\begin{aligned}\mathbf{r}'_{\theta} &= \mathbf{r}'_u u'_{\theta} + \mathbf{r}'_v v'_{\theta} \\ \mathbf{r}'_{\varphi} &= \mathbf{r}'_u u'_{\varphi} + \mathbf{r}'_v v'_{\varphi}.\end{aligned}$$

We obtain that  $\text{Span}\{\mathbf{r}'_{\theta}, \mathbf{r}'_{\varphi}\} = \text{Span}\{\mathbf{r}'_u, \mathbf{r}'_v\}$  but

$$\mathbf{r}'_{\theta} \times \mathbf{r}'_{\varphi} = \frac{\partial(u, v)}{\partial(\theta, \varphi)} \mathbf{r}'_u \times \mathbf{r}'_v.$$

The tangent plane does not depend on the particular choice of parameterization. On the other hand, we have

$$\begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = \begin{pmatrix} \mathbf{r}'_{\theta} \\ \mathbf{r}'_{\varphi} \end{pmatrix} \cdot \begin{pmatrix} \mathbf{r}'_{\theta} & \mathbf{r}'_{\varphi} \end{pmatrix} = \begin{pmatrix} u'_{\theta} & v'_{\theta} \\ u'_{\varphi} & v'_{\varphi} \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} u'_{\theta} & u'_{\varphi} \\ v'_{\theta} & v'_{\varphi} \end{pmatrix},$$

and

$$\sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} = \sqrt{EG - F^2} \left| \frac{\partial(u, v)}{\partial(\theta, \varphi)} \right|.$$

## 2.3 General surface in $\mathbf{R}^3$

Suppose  $S \subset \mathbf{R}^3$  is given implicitly by  $F(x, y, z) = 0$  where  $F : U(\subset \mathbf{R}^3) \rightarrow \mathbf{R}$  is a  $C^1$  function defined on some open domain in  $\mathbf{R}^3$ .

Suppose  $M_0 = (x_0, y_0, z_0) \in S$  such that  $\nabla F(M_0) \neq 0$ , then the tangent plane of  $S$  at  $M_0$  is given by

$$T_{M_0} S = \text{Span}\{\nabla F(M_0)\}^{\perp}.$$

If  $\nabla F(M_0) = 0$ , then  $M_0$  is called a singular point of  $S$ , for instance the vertex point of the double cone  $x^2 + y^2 - z^2 = 0$ .

**Example 2.5.** We have learned three ways of understanding the sphere of radius  $R$  in  $\mathbf{R}^3$ .

**Solution.** 1. *Parametric equation using longitude and latitude.*

2. Using implicit function: at  $\mathbf{r} = (x, y, z) \in S$ , we have an normal vector  $\nabla F = (2x, 2y, 2z)$  and so the unit normal can be given as  $\mathbf{n} = \frac{\nabla F}{|\nabla F|} = \frac{\mathbf{r}}{R}$ .
3. Parametric equation using Cartesian coordinate, for instance  $\mathbf{r}(x, y) = (x, y, \sqrt{R^2 - x^2 - y^2})$  for the north hemisphere. Then

$$\begin{aligned}\mathbf{r}'_x &= (1, 0, -\frac{x}{z}) \\ \mathbf{r}'_y &= (0, 1, -\frac{y}{z}) \\ \mathbf{r}'_x \times \mathbf{r}'_y &= \left(\frac{x}{z}, \frac{y}{z}, 1\right) \\ \mathbf{n} &= \frac{(x, y, z)}{\sqrt{x^2 + y^2 + z^2}} = \frac{\mathbf{r}}{R}.\end{aligned}$$

By studying how  $\mathbf{n}$  varies on the surface, we obtain the notion of “Gaussian curvature”, generalizing the notion of curvature for space curve.

Notice that the whole sphere can be covered by the graph of six implicit functions  $x = \pm\sqrt{R^2 - y^2 - z^2}$ ,  $y = \pm\sqrt{R^2 - z^2 - x^2}$ ,  $z = \pm\sqrt{R^2 - x^2 - y^2}$ .

For a implicit space curve given by

$$\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$$

where  $F, G$  are two  $C^1$  function. If  $\nabla F(M_0) \times \nabla G(M_0) \neq \mathbf{0}$ , then  $\mathbf{l} = \nabla F(M_0) \times \nabla G(M_0) = \left(\frac{\partial(F,G)}{\partial(y,z)}, \frac{\partial(F,G)}{\partial(z,x)}, \frac{\partial(F,G)}{\partial(x,y)}\right)(M_0)$  is the tangent vector to the intersection curve at  $M_0$ .

### 3 Differentiation of vector fields

Many physical quantities are described by “vector fields” which is the prescription of a tangent vector at each point of the space in consideration. For instance, electric field, magnetic field, force field and velocity field in fluid mechanics.

A vector field is a vector-valued function  $\mathbf{v} : D(\subset \mathbf{R}^3) \rightarrow \mathbf{R}^3$ . In Cartesian coordinate, it can be written as

$$\mathbf{v}(x, y, z) = P(x, y, z)\mathbf{i} + Q(x, y, z)\mathbf{j} + R(x, y, z)\mathbf{k},$$

where  $P, Q, R$  are called the component function of  $\mathbf{v}$ .

**Example 3.1.** •  $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$  is the position vector field of the space;

- $\mathbf{F} = -G \frac{Mm}{r^3} \mathbf{r}$  describes the gravity force of a point mass  $M$  (posited at the origin) acting on a point mass  $m$  at the position  $\mathbf{r}$ ;
- $\mathbf{v} = -y\mathbf{i} + x\mathbf{j}$  is a vortex vector field in the plane.

We should notice that a function defined on the space is called *scalar field* in contrast to *vector field*, for instance, the temperature, pressure, density....

As vector-valued function

$$d\mathbf{v} = \frac{\partial \mathbf{v}}{\partial x} dx + \frac{\partial \mathbf{v}}{\partial y} dy + \frac{\partial \mathbf{v}}{\partial z} dz,$$

where for instance

$$\frac{\partial \mathbf{v}}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{\mathbf{v}(x + \Delta x, y, z) - \mathbf{v}(x, y, z)}{\Delta x} = \frac{\partial P}{\partial x} \mathbf{i} + \frac{\partial Q}{\partial x} \mathbf{j} + \frac{\partial R}{\partial x} \mathbf{k}.$$

Those  $\frac{\partial \mathbf{v}}{\partial x}, \frac{\partial \mathbf{v}}{\partial y}, \frac{\partial \mathbf{v}}{\partial z}$  are also vector field.

### 3.1 Gradient, divergence and Curl

For a scalar field  $\varphi : \mathbf{R}^3 \rightarrow \mathbf{R}$ ,

$$d\varphi = \frac{\partial \varphi}{\partial x} dx + \frac{\partial \varphi}{\partial y} dy + \frac{\partial \varphi}{\partial z} dz = \nabla \varphi \cdot d\mathbf{r}. \quad (3.1)$$

The vector field

$$\nabla \varphi = \frac{\partial \varphi}{\partial x} \mathbf{i} + \frac{\partial \varphi}{\partial y} \mathbf{j} + \frac{\partial \varphi}{\partial z} \mathbf{k} \quad (3.2)$$

is called the *gradient vector field* of  $\varphi$ . We call

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

the *Hamilton operator* or *Nabla operator*. Notice that this operator both has the vector part and the differentiation part.

We can also define

$$\nabla \cdot \mathbf{v} = \left( \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \cdot (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \quad (3.3)$$

for any differentiable vector field  $\mathbf{v} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ . This is called the *divergence* of the vector field  $\mathbf{v}$ .

Beside “dot product”, we can also define

$$\begin{aligned} \nabla \times \mathbf{v} &= \left( \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) \times (P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \\ &= \left( \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left( \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \end{aligned} \quad (3.4)$$

for any differentiable vector field  $\mathbf{v} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ . This is called the *rot* or *curl* of the vector field  $\mathbf{v}$ .

The geometric and physical meaning of divergence and curl will be given in the chapter of vector field integration.

### 3.2 Nabla operator under cylindrical coordinate

Notice that the definition of Nabla operator

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

seems to depend on the particular choice of Cartesian coordinate system.

Recall the cylindrical coordinate in  $\mathbf{R}^3$  given by  $(x, y, z) = (r \cos \theta, r \sin \theta, z)$ . For any function  $u$  defined on  $\mathbf{R}^3$ , then under the  $x, y, z$  coordinate

$$\nabla u = \left( \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) u, \quad (3.5)$$

and by Chain Rule under the  $r, \theta, z$  coordinate we have

$$\begin{aligned} \nabla u &= \mathbf{i} \left( \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} + \frac{\partial z}{\partial x} \frac{\partial}{\partial z} \right) u \\ &\quad + \mathbf{j} \left( \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta} + \frac{\partial z}{\partial y} \frac{\partial}{\partial z} \right) u \\ &\quad + \mathbf{k} \left( \frac{\partial r}{\partial z} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial z} \frac{\partial}{\partial \theta} + \frac{\partial z}{\partial z} \frac{\partial}{\partial z} \right) u \\ &= \left[ (\cos \theta \mathbf{i} + \sin \theta \mathbf{j}) \frac{\partial}{\partial r} + \frac{1}{r} (-\sin \theta \mathbf{i} + \cos \theta \mathbf{j}) \frac{\partial}{\partial \theta} + \mathbf{k} \frac{\partial}{\partial z} \right] u \\ &= \left( \mathbf{e}_r \frac{\partial}{\partial r} + \frac{1}{r} \mathbf{e}_\theta \frac{\partial}{\partial \theta} + \mathbf{e}_z \frac{\partial}{\partial z} \right) u. \end{aligned}$$

Therefore, the Nabla operator takes the form

$$\nabla = \mathbf{e}_r \frac{\partial}{\partial r} + \frac{1}{r} \mathbf{e}_\theta \frac{\partial}{\partial \theta} + \mathbf{e}_z \frac{\partial}{\partial z}$$

in the cylindrical coordinate, where  $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z\}$  forms moving frame for  $T_{\mathbf{x}}\mathbf{R}^3$ .

We list the properties of Nabla operator below:

$$\begin{aligned} \nabla(fg) &= (\nabla f)g + f(\nabla g) \\ \nabla \cdot (f\mathbf{u}) &= f\nabla \cdot \mathbf{u} + \nabla f \cdot \mathbf{u} \\ \nabla \cdot (\mathbf{u} + \mathbf{v}) &= \nabla \cdot \mathbf{u} + \nabla \cdot \mathbf{v} \\ \nabla \times (f\mathbf{u}) &= \nabla f \times \mathbf{u} + f\nabla \times \mathbf{u} \\ \nabla \times (\mathbf{u} + \mathbf{v}) &= \nabla \times \mathbf{u} + \nabla \times \mathbf{v} \\ \nabla \cdot (\mathbf{a} \times \mathbf{b}) &= \mathbf{b} \cdot \nabla \times \mathbf{a} - \mathbf{a} \cdot \nabla \times \mathbf{b}. \\ \nabla \times \nabla \phi &= \mathbf{0}, \quad \forall \phi \in C^2(\mathbf{R}^3) \\ \nabla \cdot \nabla \times \mathbf{v} &= \mathbf{0}, \quad \forall \mathbf{v} \in C^2(\mathbf{R}^3, \mathbf{R}^3). \end{aligned}$$

The last two equations say that “any gradient vector field must be curl free, and any curl vector field is divergence free”.

Finally, we need to mention the *Laplace operator*: for any  $\phi \in C^2$ , we define

$$\Delta\phi = \nabla \cdot \nabla\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}. \quad (3.6)$$

We can compute the Laplace operator under other coordinates, such as spherical coordinate or cylindrical coordinate by direct computation.

## 4 Differential Forms

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## 5 Integration of Multivariate function

### 5.1 Area of subset in $\mathbf{R}^2$

Suppose  $D \subset \mathbf{R}^2$  be a subset contained in  $[a, b] \times [c, d]$ . Let  $T : a = x_0 < x_1 < \cdots < x_n = b, c = y_0 < y_1 < \cdots < y_m$  be a rectangular partition of  $[a, b] \times [c, d]$ . Denote  $R_{ij}$  the  $(i, j)$ -th rectangle  $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$ , and  $\|T\| = \max\{\Delta x_i, \Delta y_j | i = 1, \cdots, n; j = 1, \cdots, m\}$  the modulus of the partition.

We call

$$\begin{aligned} \sigma_T^+(D) &= \sum_{R_{ij} \cap D \neq \emptyset} \sigma(R_{ij}) \\ \sigma_T^-(D) &= \sum_{R_{ij} \subset D} \sigma(R_{ij}) \end{aligned}$$

the *outer area* and *inner area* of  $D$  with respect to  $T$  respectively. If

$$\lim_{\|T\| \rightarrow 0} \sigma_T^+(D) = \lim_{\|T\| \rightarrow 0} \sigma_T^-(D),$$

then we say  $D$  is *Jordan measurable*, and the limit denote by  $\sigma(D)$  is called the *Jordan measure* of  $D$ .

It is not difficult to see that

$$0 \leq \sigma_T^+(D) - \sigma_T^-(D) \leq \sigma_T^+(\partial D) = \sigma_T^+(D) - \sigma_T^-(\overset{\circ}{D}).$$

This means that if  $\partial D$  has Jordan measure 0, then  $D$  is Jordan measurable. In Mathematical Analysis III we will show the converse, i.e. the following theorem holds.

**Theorem 5.1.**  $D$  is Jordan measurable iff  $\partial D$  has Jordan measure 0.

**Example 5.2.**

- Let  $D = \{(x, f(x)) | x \in [a, b]\}$  be the graph of the continuous function  $f$  on bounded closed interval  $[a, b]$ , then  $D$  has Jordan measure 0;

- If the boundary  $\partial D$  can be represented by finitely many curves which can be written as the graph of continuous function, then  $D$  is Jordan measurable;
- We can restrict to  $n = 1$  to define Jordan measure for subset of  $\mathbf{R}$ . The set  $[0, 1] \cap \mathbf{Q}$  is not Jordan measurable, and similarly the set  $[0, 1]^2 \cap \mathbf{Q}^2$  is not Jordan measurable.

Recall the area of the set under a one variable function  $y = f(x)$  on the interval  $[a, b]$ . The Darboux upper sum and Darboux lower sum there can be interpreted as the outer area and inner area of  $D$  here, and

$$\sigma_{T'}^+(D) - \sigma_{T'}^-(D) = \sum_{i=1}^n \omega_i \Delta x_i$$

where  $T'$  is the rectangular partition of  $D$  induced by the partition  $T$  of the interval  $[a, b]$  (i.e. we take the vertical partition of  $[0, \sup f]$  using the point set

$$\left\{ \sup_{[x_{i-1}, x_i]} f, \inf_{[x_{i-1}, x_i]} f \mid i = 1, 2, \dots, n \right\}.$$

Thus the Riemann integrability of  $f$  is equivalent to the Jordan measurability of  $D$ , and their values are equal in the case of measurable. It is not difficult to believe (will be proved next semester) that Jordan measure satisfies the “expected properties” as usual “area” for a certain reasonable class of point sets.

However, we also need to see that simple set like  $\mathbf{Q}^2$  are not Jordan measurable. That’s one of the starting point in the pursuit of more general notion of measures, such as **Lebesgue measure, Hausdorff measure, Radon measure.....**

## 5.2 Concept and properties of double integral

Let  $D \subset \mathbf{R}^2$  be a measurable subset, a partition of  $D$  means a division of  $D$  into  $n$  non overlapping measurable subset  $D_i$  (i.e.  $\overset{\circ}{D}_i \cap \overset{\circ}{D}_j = \emptyset$ ) and denote

$$\|T\| = \max_{1 \leq i \leq n} \text{diam } D_i.$$

Take a point  $(\xi_i, \eta_i) \in D_i$  and form the Riemann sum

$$S(T, \{(\xi_i, \eta_i)\}; f) = \sum_{i=1}^n f(\xi_i, \eta_i) \sigma(D_i). \quad (5.1)$$

**Definition 5.3.** If there exists  $A$  satisfying:  $\forall \varepsilon > 0, \exists \delta > 0$  such that for any partition  $T$  of  $D$  with  $\|T\| < \delta$  we have

$$|S(T, \{(\xi_i, \eta_i)\}; f) - A| < \varepsilon$$

for any choice of point  $(\xi_i, \eta_i) \in D_i$ , then we say  $f$  is Riemann integrable on  $D$ , and denote

$$A = \iint_D f(x, y) d\sigma.$$

Here  $A$  is called the integration of  $f$ ,  $D$  is called the integrating region,  $f(x, y)$  is called the integrand,  $d\sigma$  is called area element.

Geometrically, the Riemann sum is approximately the volume of region under a surface of graph of  $f$ . The more refined the partition, the more precise this sum is relative to the true volume.

**Theorem 5.4.** *Integrable function must be bounded.*

Try to use definition to verify the two dimensional Dirichlet function is not Riemann integrable.

Similarly to  $n = 1$  case, we can define

$$\begin{aligned}\overline{S}(T, f) &= \sum_{i=1}^n M_i \sigma(D_i), \\ \underline{S}(T, f) &= \sum_{i=1}^n m_i \sigma(D_i) \\ \omega(T, f) &= \overline{S}(T, f) - \underline{S}(T, f)\end{aligned}\tag{5.2}$$

the Darboux upper sum, Darboux lower sum and Darboux oscillation of  $f$  with respect to the partition  $D$ . The infimum of Darboux upper sum over all partition is called the Darboux upper integral, denoted as  $\overline{I}(f)$ , and similarly we can define  $\underline{I}(f)$ .

**Theorem 5.5.** *Let  $f$  be a bounded function on a measurable set  $D$ .*

1.  $f$  is Riemann integrable iff  $\overline{I}(f) = \underline{I}(f)$ ;
2. If  $f$  is continuous on  $D$ , then  $f$  is integrable. If the discontinuity of  $f$  lies in a subset of Jordan measure 0, then  $f$  is also integrable;
3. If  $f$  and  $g$  only differs on a subset of Jordan measure zero, then  $f$  is integrable iff  $g$  is integrable, and when integrable

$$\iint_D f d\sigma = \iint_D g d\sigma.$$

The proofs are left as exercises. The two dimensional Dirichlet function is not Riemann integrable as  $\overline{I}(f) - \underline{I}(f) = 1$ .

**Theorem 5.6.** *Let  $D$  be a measurable set and  $f, g$  are integrable functions on  $D$ .*

1. For any  $c_1, c_2 \in \mathbf{R}$ ,  $c_1 f + c_2 g$  is integrable and

$$\iint_D (c_1 f + c_2 g) d\sigma = c_1 \iint_D f d\sigma + c_2 \iint_D g d\sigma.$$

2. Suppose  $D = D_1 \cup D_2$  where  $D_1, D_2$  are measurable and  $\mathring{D}_1 \cap \mathring{D}_2 = \emptyset$ . If  $f$  is integrable on  $D_1$  and  $D_2$ , then

$$\iint_D f d\sigma = \iint_{D_1} f d\sigma + \iint_{D_2} f d\sigma.$$

3. The product  $fg$  is also integrable on  $D$  with

$$\left| \iint_D fg d\sigma \right|^2 \leq \iint_D f^2 d\sigma \iint_D g^2 d\sigma.$$

4. If  $f \geq g$  on  $D$ , then

$$\iint_D f d\sigma \geq \iint_D g d\sigma.$$

In particular, if  $m \leq f(x, y) \leq M$  for  $(x, y) \in D$ , then

$$m\sigma(D) \leq \iint_D f d\sigma \leq M\sigma(D).$$

5. The function  $|f|$  is also integrable with

$$\left| \iint_D f d\sigma \right| \leq \iint_D |f| d\sigma.$$

6. (Mean Value Theorem.) If  $D$  is a connected closed measurable set and  $f$  is continuous on  $D$ , then there exists  $P \in D$  such that

$$\iint_D f d\sigma = f(P)\sigma(D).$$

For item 6, we can use the Theorem that the intermediate value property for continuous function on connected set.

### 5.3 Computation of double integral

The Newton-Leibniz Formula/ Fundamental Theorem of Calculus is a powerful tool in the computation of integration for  $n = 1$ . Do we have a similar formula for  $n = 2$ ?

Let  $D = [a, b] \times [c, d]$  and  $f$  be integrable on  $D$ . Take the partition of  $D$  as  $T = T_x \times T_y$  where

$$\begin{aligned} T_x : a = x_0 < x_1 < \cdots < x_n = b \\ T_y : c = y_0 < y_1 < \cdots < y_m = d. \end{aligned}$$

This gives Riemann sum

$$S(T, \{M_{ij}\}; f) = \sum_{i=1}^n \sum_{j=1}^m f(M_{ij})\sigma(R_{ij}) = \sum_{i=1}^n \sum_{j=1}^m f(M_{ij})\Delta x_i \Delta y_j.$$



If we take the particular choice  $M_{ij} = (\xi_i, \eta_j)$  and let  $\max \Delta x_i$  and  $\max \Delta y_j$  go to 0 step by step, then the above limit can be seen equal to

$$\int_c^d \left( \int_a^b f(x, y) dx \right) dy.$$

The above is not rigorous proof since it involves iterative limit.

**Theorem 5.7 (Iterative integration formula).** Suppose  $f$  is integrable on  $D = [a, b] \times [c, d]$ .

1. If for any  $y \in [c, d]$ ,  $f(\cdot, y)$  is Riemann integrable with respect to  $x \in [a, b]$ , then the function  $\varphi(y) := \int_a^b f(x, y) dx$  of  $y \in [c, d]$  is integrable on  $[c, d]$ , and

$$\int_c^d \varphi(y) dy = \int_c^d \left( \int_a^b f(x, y) dx \right) dy = \iint_D f(x, y) d\sigma.$$

2. If for any  $x \in [a, b]$ ,  $f(x, \cdot)$  is Riemann integrable with respect to  $y \in [c, d]$ , then the function  $\psi(x) := \int_c^d f(x, y) dy$  is integrable on  $[a, b]$ , and

$$\int_a^b \psi(x) dx = \int_a^b \left( \int_c^d f(x, y) dy \right) dx = \iint_D f(x, y) d\sigma.$$

3. If the conditions of item 1 and 2 hold simultaneously, then

$$\iint_D f(x, y) d\sigma = \int_c^d dy \int_a^b f(x, y) dx = \int_a^b dx \int_c^d f(x, y) dy.$$

*Proof.* By the integrability of  $f(x, y)$  on  $[a, b] \times [c, d]$ , for any  $\varepsilon > 0$ , there exists  $\delta = \delta(\varepsilon) > 0$  such that for any two dimensional rectangular partition  $T$  with  $\|T\| < \delta$ ,

$$\left| \sum_{1 \leq i \leq n, 1 \leq j \leq m} f(M_{ij}) \Delta x_i \Delta y_j - A \right| < \varepsilon. \quad (5.3)$$

For the above  $\varepsilon$ , take arbitrary partition  $T_x : a = x_0 < x_1 < \cdots < x_n = b$  with  $\|T_x\| < \delta$  (which is the above  $\delta$ ). Then for any sampling points  $\{\xi_i\}_{i=1, \dots, n}$ , we can choose  $\delta' = \delta'(\xi_1, \dots, \xi_n, \varepsilon) > 0$  such that

$$\left| \psi(\xi_i) - \sum_{j=1}^n f(\xi_i, \eta_j) \Delta y_j \right| < \varepsilon, \quad (5.4)$$

for any partition  $T_y : c = y_0 < y_1 < \cdots < y_m = d$  and  $\eta_j \in [y_{j-1}, y_j]$  and any  $i = 1, 2, \dots, n$  according to the integrability of  $f(\xi_i, \cdot)$  on  $[c, d]$ .

Now let us fix one particular partition  $T_y$  with  $\|T_y\| < \delta'' = \min\{\delta', \delta\}$ , then we can use (5.3) and (5.4) (with the particular sampling  $\{(\xi_i, y_j)\}$ ) to conclude that

$$\begin{aligned} \sum_{j=1}^n f(\xi_i, y_j) \Delta y_j - \varepsilon &< \psi(\xi_i) < \sum_{j=1}^n f(\xi_i, y_j) \Delta y_j + \varepsilon \\ A - \varepsilon &< \sum_{1 \leq i \leq n, 1 \leq j \leq m} f(\xi_i, y_j) \Delta y_j \Delta x_i < A + \varepsilon. \end{aligned} \quad (5.5)$$

Combining the two inequalities, we conclude that

$$A - \varepsilon - \sum_{i=1}^n \varepsilon \Delta x_i < \sum_{i=1}^n \psi(\xi_i) \Delta x_i < A + \varepsilon + \sum_{i=1}^n \varepsilon \Delta x_i, \quad (5.6)$$

i.e.

$$\left| \sum_{i=1}^n \psi(\xi_i) \Delta x_i - \iint_{[a,b] \times [c,d]} f(x, y) dx dy \right| < (1 + b - a) \varepsilon,$$

provided the initial assumption on  $T_x$  that  $\|T_x\| < \delta$  holds. This verifies that  $\varphi$  is integrable on  $[a, b]$ . Moreover, the formula

$$\int_a^b \psi(x) dx = \iint_{[a,b] \times [c,d]} f(x, y) d\sigma$$

holds. □

**Remark 5.8.** Notice that  $\delta', \delta''$  are only intermediate variables which we use only to get the final inequality. The validity of the final inequality does not depend on those parameters.  $\delta', \delta''$  might depend on the sampling points  $\{\xi_i\}$ , but the final  $\delta$  does depend only on  $\varepsilon$ .

A similar phenomenon happens when we prove the uniform limit  $f$  of  $f_n \in C[a, b]$  is continuous. There, we use an intermediate  $N$  and the continuity of  $f_N$  to get an intermediate modulus of continuity  $\delta$ . That  $\delta$  is also good for  $f$ .

Geometrically, this means the volume of a bread can be computed by slicing it into very thin pieces and then compute the sum of the volume of all slices.

By the reason of this theorem, we sometimes write  $d\sigma = dx dy$ .

**Remark 5.9.** Let  $f$  be continuous on  $[a, b] \times [c, d]$ , then

$$F(x, y) := \iint_{[a,x] \times [c,y]} f(s, t) ds dt$$

defines a continuous function on  $[a, b] \times [c, d]$  such that

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x} = f.$$

However, this will not be treated as proper generalization of Newton-Leibniz's Formula as it involves second order derivatives of  $F$ .

**Example 5.10.** Find  $\iint_{[0,1] \times [0,1]} e^{x+y} dx dy$ .

**Solution.**

$$\int_{[0,1] \times [0,1]} e^{x+y} dx dy = \int_0^1 dy \int_0^1 e^{x+y} dx = \int_0^1 (e^{1+y} - e^y) dy = (e - 1)^2.$$

**Example 5.11.** Calculate  $\iint_{[0,\pi] \times [0,1]} x \cos xy dx dy$ .

**Solution.**

$$\iint_{[0,\pi] \times [0,1]} x \cos xy dx dy = \int_0^\pi dx \int_0^1 x \cos xy dy = \int_0^\pi \sin xy \Big|_{y=0}^{y=1} dx = \int_0^\pi \sin x dx.$$

If we try the other order, then we get

$$\begin{aligned} \int_0^1 dy \int_0^\pi x \cos xy dx &= \int_0^1 dy \left( \int_0^\pi x \frac{\sin xy}{y} \Big|_{x=0}^{x=\pi} - \int_0^\pi \frac{\sin xy}{y} dx \right) \\ &= \int_0^1 dy \left( \pi \frac{\sin \pi y}{y} + \frac{\cos \pi y - 1}{y^2} \right) dy \\ &=? \end{aligned}$$

for which the antiderivative is difficult to find. Notice that  $y = 0$  is the removable singularity of the integrand function in the final step.

We can actually generalize the iterative integration formula to more general regions.

**Theorem 5.12.**

1. Let  $D = \{(x, y) | y_1(x) \leq y \leq y_2(x), a \leq x \leq b\}$  where  $y_1, y_2$  are continuous on  $[a, b]$ , and suppose  $f$  is integrable on  $D$ .

If for all  $x \in [a, b]$ ,  $f(x, \cdot)$  is integrable on  $[y_1(x), y_2(x)]$  with respect to  $y$ , then the function  $\varphi(x) = \int_{y_1(x)}^{y_2(x)} f(x, y) dy$  of the variable  $x \in [a, b]$  is Riemann integrable, and

$$\int_a^b \varphi(x) dx = \int_a^b dx \int_{y_1(x)}^{y_2(x)} f(x, y) dy = \iint_D f(x, y) dx dy.$$

2. Let  $D = \{(x, y) | x_1(y) \leq x \leq x_2(y), c \leq y \leq d\}$  where  $x_1, x_2$  are continuous on  $[c, d]$ . If for all  $y \in [c, d]$ ,  $f(\cdot, y)$  is integrable on  $[x_1(y), x_2(y)]$  then

$$\int_c^d dy \int_{x_1(y)}^{x_2(y)} f(x, y) dx = \iint_D f(x, y) dx dy.$$

If the region can be divided into the non-overlapping union of several region of the shapes described in the theorem, then we can use the additivity of double integral to compute the integral.

*Proof.* Suppose the range of  $y_1(x)$  and  $y_2(x)$  are both bounded by  $[c, d]$ . Extend the function  $f$  to  $[a, b] \times [c, d]$  by 0, i.e. define

$$f^*(x, y) = \begin{cases} f(x, y), & (x, y) \in D, \\ 0, & (x, y) \notin D. \end{cases}$$

Then  $[a, b] \times [c, d] = D \cup ([a, b] \times [c, d] \setminus D)$  is a decomposition into measurable sets, and moreover  $f^*$  is both integrable on  $D$  and  $[a, b] \times [c, d] \setminus D$ . This implies that  $f^*$  is integrable on  $[a, b] \times [c, d]$ . Using the proved theorem,

$$\varphi^*(x) = \int_c^d f^*(x, y) dy = \int_{y_1(x)}^{y_2(x)} f(x, y) dy = \varphi(x)$$

is integrable about  $x \in [a, b]$  and

$$\begin{aligned} \int_a^b \varphi(x) dx &= \int_a^b dx \int_c^d f^*(x, y) dy \\ &= \int_a^b dx \int_{y_1(x)}^{y_2(x)} f(x, y) dy \\ &= \iint_{[a, b] \times [c, d]} f^*(x, y) dx dy \\ &= \iint_D f(x, y) dx dy. \end{aligned}$$

□

**Example 5.13.** Suppose  $f(x, y)$  is continuous, prove that

$$\int_a^b dx \int_a^x f(x, y) dy = \int_a^b dy \int_y^b f(x, y) dx.$$

*Proof.* Use the iterative integration formula, we obtain that

$$\int_a^b dx \int_a^x f(x, y) dy = \iint_{a \leq x \leq b, a \leq y \leq x} f(x, y) dx dy = \int_a^b dy \int_y^b f(x, y) dx.$$

□

**Example 5.14.** Let  $D$  be enclosed by  $y = 0, x = a$  and  $y = \frac{b}{a}x$ . Compute  $\iint_D x^2 y^2 dx dy$ .

**Solution.**

$$\begin{aligned} \int_D x^2 y^2 dx dy &= \int_0^a dx \int_0^{\frac{b}{a}x} x^2 y^2 dy \\ &= \int_0^a x^2 \cdot \frac{1}{3} \left( \frac{b}{a}x \right)^3 dx \\ &= \frac{1}{18} a^3 b^3. \end{aligned}$$

**Example 5.15.** Let  $D$  be enclosed by  $y = a, y = 3a, y = x$  and  $y = x + a$ . Find the integral  $\iint_D (x^2 + y^2) dx dy$ .

**Solution.** If we integrate first with respect to  $y$ , then we need to decompose the region into three piece  $0 \leq x \leq a, a \leq x \leq 2a, 2a \leq x \leq 3a$ . We integrate with respect to  $x$  first, then one region is enough.

$$\begin{aligned} \int_D (x^2 + y^2) dx dy &= \int_a^{3a} dy \int_{y-a}^y (x^2 + y^2) dx \\ &= \int_a^{3a} \left( \frac{x^3}{3} + y^2 x \right) \Big|_{y-a}^y dy \\ &= \int_a^{3a} \left( ay^2 + \frac{y^3}{3} - \frac{(y-a)^3}{3} \right) dy \\ &= 14a^4. \end{aligned}$$

**Example 5.16.** Find the iterative integration

$$\int_1^2 y dy \int_y^2 \frac{\sin x}{x^2 - 1} dx.$$

**Solution.** We have to understand this integration in the mean of improper integral since at  $y = 1$ , the interior function  $\int_y^2 \frac{\sin x}{x^2 - 1} dx = +\infty$ . Thus we understand the whole integral as

$$\lim_{\delta \rightarrow 0} \int_{1+\delta}^2 y dy \int_y^2 \frac{\sin x}{x^2 - 1} dx.$$

Before taking limit, by the proved theorem, this iterative integral is equal to the integral

$$\begin{aligned} \iint_{1+\delta \leq y \leq 2, 1+\delta \leq x \leq y} \frac{\sin x}{x^2 - 1} y dx dy &= \int_{1+\delta}^2 \frac{\sin x}{x^2 - 1} dx \int_{1+\delta}^x y dy \\ &= \int_{1+\delta}^2 \frac{\sin x}{x^2 - 1} (x^2 - 1 - 2\delta - \delta^2) dx. \end{aligned} \quad (5.7)$$

Notice that

$$\left| \int_{1+\delta}^2 \frac{\sin x}{x^2 - 1} dx \right| \leq \frac{1}{2} \int_{1+\delta}^2 \frac{1}{x - 1} dx = \frac{1}{2} \log \frac{1}{\delta}.$$

The consequence is that

$$\lim_{\delta \rightarrow 0} \int_{1+\delta}^2 y dy \int_y^2 \frac{\sin x}{x^2 - 1} dx = \lim_{\delta \rightarrow 0} \int_{1+\delta}^2 \sin x dx = \int_1^2 \sin x dx.$$

**Example 5.17.** Find  $\iint_{|x|+|y| \leq 1} x^2 y^2 dx dy$ .

**Solution.** By symmetry of the region and the function, we get that

$$\begin{aligned}\iint_{|x|+|y|\leq 1} x^2 y^2 dx dy &= 4 \int_0^1 dx \int_0^{1-x} x^2 y^2 dy \\ &= \frac{4}{3} \int_0^1 x^2 (1-x)^3 dx \\ &= \frac{1}{45}.\end{aligned}$$

**Example 5.18.** Find the volume of the body enclosed by two cylinders  $x^2 + y^2 \leq a^2$  and  $x^2 + z^2 \leq a^2$ . 牟盒方盖

**Solution.** The enclosed region is enclosed by the graph of the function  $z = \sqrt{a^2 - x^2}$  and  $z = -\sqrt{a^2 - x^2}$  restricted to the region  $x^2 + y^2 \leq a^2$ . The volume is therefore

$$\begin{aligned}V &= \iint_{x^2+y^2\leq a^2} 2\sqrt{a^2-x^2} dx dy \\ &= 8 \int_0^a \sqrt{a^2-x^2} dx \int_0^{\sqrt{a^2-x^2}} dy \\ &= 8 \int_0^a (a^2-x^2) dx \\ &= \frac{16}{3} a^3.\end{aligned}$$

If the horizontal cylinder is given by  $x^2 + z^2 \leq b^2$ , what is the enclosed volume. The main problem is about the integration region. For  $b < a$ , we have to consider the region  $\{x^2 + y^2 \leq a^2\} \cap \{x^2 \leq b^2\}$ , and the volume is then

$$\int_{x^2+y^2\leq a^2, x^2\leq b^2} 2\sqrt{b^2-x^2} dx dy.$$

## 6 Change of Variables for computation

Consider for instance the double integral

$$\iint_{x^2+y^2\leq 1, x\geq 0, y\geq 0} \sqrt{x^2+y^2} dx dy = \int_0^1 dx \int_0^{\sqrt{1-x^2}} \sqrt{x^2+y^2} dy.$$

However, the integrand now becomes

$$\frac{1}{2} \left( \sqrt{1-x^2} + x^2 \log \frac{1+\sqrt{1-x^2}}{x} \right)$$

which is difficult to integrate on  $[0, 1]$ .

However, if we go back to the definition of double integral and do the partition of the region in terms of rectangular partition in  $(r, \theta)$  coordinate:  $T : 0 = r_0 < r_1 < \dots < r_m = 1, \theta_0 = 0 < \theta_1 < \dots < \theta_n = \frac{\pi}{2}$ , then the Riemann sum becomes

$$\sum_{i,j} f(M_{ij}) \sigma(D_{ij}) = \sum_{i,j} r_i \cdot \frac{1}{2} (r_i^2 - r_{i-1}^2) (\theta_j - \theta_{j-1}) \sim \sum_{i,j} r_i^2 \Delta r_i \Delta \theta_j.$$

It is easily to see that the limit of such sequence of Riemann sums is

$$\int_0^1 r^2 dr \cdot \int_0^{\frac{\pi}{2}} d\theta = \frac{\pi}{6}.$$

This means that using different partitions, the computation of the integral could be probably significantly simplified.

Suppose  $D' \rightarrow D$  by sending  $(u, v)$  to  $(x, y) = (x(u, v), y(u, v))$  is a coordinate transformation, and we form the Riemann sum of  $f : D \rightarrow \mathbf{R}$  by dividing  $D$  into regions resulted from rectangular partition of  $D'$ .

Let  $[u, u + \Delta u] \times [v, v + \Delta v]$  be a rectangle in  $D'$ , then the image under the coordinate transformation is a curvilinear rectangle  $M_1 M_2 M_3 M_4$  in  $D$  whose area is approximately the area of the parallelogram  $M_1 M_2 M_3 M_4$ , i.e.

$$\begin{aligned} \left| \overrightarrow{M_1 M_2} \times \overrightarrow{M_1 M_4} \right| &= \left| (x'_u|_{(u,v)} \Delta u + o(\Delta u)) (y'_v|_{(u,v)} \Delta v + o(\Delta v)) \right. \\ &\quad \left. - (x'_v|_{(u,v)} \Delta v + o(\Delta v)) (y'_u|_{(u,v)} \Delta u + o(\Delta u)) \right| \quad (6.1) \\ &= \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \Delta u \Delta v + o(\Delta u \cdot \Delta v) \end{aligned}$$

2

Now let us consider the partition  $T : u_0 < u_1 < \dots < u_n, v_0 < v_1 < \dots < v_m$  of the rectangle which  $D'$  lies in. Let  $\{D'_1, \dots, D'_N\}$  be the rectangles inside  $D'$  and let  $\{D_1, \dots, D_N\}$  be the curvilinear rectangles under the coordinate transformation:

$$\sum_{k=1}^N f(M_k) \sigma(D_k) \sim \sum_{k=1}^N f(M'_k) \left| \frac{\partial(x, y)}{\partial(u, v)} (M'_k) \right| \sigma(D'_k)$$

Therefore, the limit of Riemann sum is

$$\iint_D f(x, y) dx dy = \iint_{D'} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

---

<sup>2</sup>There are two types of errors in doing this approximation: one caused by replacing the quadrilateral  $M_1 M_2 M_3 M_4$  by the parallelogram  $M_1 M_2 M_4$ , and the other caused by replacing each curvilinear edge by straight lines. Each step introduces an extra error of order  $o(\Delta u \Delta v)$ . The precise formulation is left as homework.

**Theorem 6.1 (Substitution Formula).** Let  $D, D'$  be two bounded open domain on  $\mathbf{R}^2$  whose boundary are consisting of finitely many piecewisely  $C^1$  curve. Suppose  $x = x(u, v), y = y(u, v)$  is a  $C^1$  map from  $D'$  to  $D$  which is one to one, and  $f(x, y)$  is continuous on  $D$ , moreover assume

$$\frac{\partial(x, y)}{\partial(u, v)} \neq 0,$$

then

$$\iint_D f(x, y) dx dy = \iint_{D'} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

The proof is quite involved, and left to next semester.

The theorem can be memorized by memorizing the transformation law for area elements:

$$dx dy = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv, \quad (6.2)$$

in analogy with  $dx = \varphi'(u) du$  in one-variable substitution formula.

**Example 6.2.** Find the volume of the ellipsoid  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$ .

**Solution.** The ellipsoid can be viewed as the body between the graph of the function  $z = c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$  and the function  $z = -c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$  defined on the closed region  $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$ . Therefore, the volume is

$$\begin{aligned} \iint_{\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1} 2c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} dx dy &=_{x=ar \cos \theta, y=br \sin \theta} 2c \iint_{r \in [0, 1], \theta \in [0, 2\pi]} \sqrt{1 - r^2} ab r dr d\theta \\ &= 2abc \int_0^{2\pi} d\theta \int_0^1 r \sqrt{1 - r^2} dr \\ &= \frac{4\pi}{3} abc. \end{aligned}$$

**Example 6.3.** Find the volume of the part of the ball  $x^2 + y^2 + z^2 \leq a^2$  cut by the cylinder  $x^2 + y^2 = ay$ .

**Solution.** The volume is

$$\begin{aligned} \iint_{x^2 + y^2 \leq ay} 2\sqrt{a^2 - x^2 - y^2} dx dy &= \iint_{\theta \in [0, \pi], r \leq a \sin \theta} 2\sqrt{a^2 - r^2} r dr d\theta \\ &= \frac{2}{3} \int_0^\pi a^3 (1 - \cos^3 \theta) d\theta \\ &= \left( \frac{2\pi}{3} - \frac{8}{9} \right) a^3. \end{aligned}$$

**Example 6.4.** Find the area enclosed by the leiminiscate  $(x^2 + y^2)^2 = 2a^2(x^2 - y^2)$ .



**Solution.** By the homogeneous property of the defining equation for the curve, it seems easier to understand it in polar coordinate

$$r^2 = 2a^2 \cos 2\theta.$$

This means the curve is confined to the region  $\theta \in [0, \frac{\pi}{4}] \cup [\frac{3\pi}{4}, \frac{5\pi}{4}] \cup [\frac{7\pi}{4}, 2\pi]$ . We can sketch its graph as below:

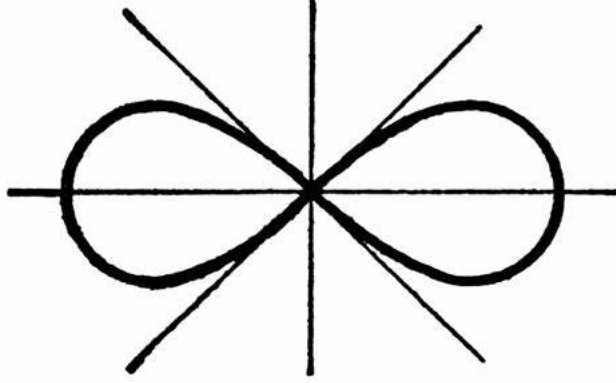


Figure 1: Bernoulli Lemniscate

It is therefore clear that the region enclosed by this lemniscate is defined as  $D' : r^2 \leq 2a^2 \cos 2\theta$ . Therefore, the area of the region in  $xOy$  coordinate is<sup>3</sup>

$$\iint_D dxdy = \iint_{D'} r dr d\theta = 4 \int_0^{\frac{\pi}{4}} d\theta \int_0^{\sqrt{2a^2 \cos 2\theta}} r dr = 4a^2 \int_0^{\frac{\pi}{4}} \cos 2\theta d\theta = 2a^2.$$

**Example 6.5.** Calculate  $\iint_D (\sqrt{x} + \sqrt{y}) dxdy$ , where  $D$  is enclosed by  $x = 0, y = 0$  and  $\sqrt{x} + \sqrt{y} = 1$ .

**Solution.** Let us try two ways of substituting.

1. Method 1: the  $(x, y) = (r^2 \cos^4 t, r^2 \sin^4 t)$ . This is a  $C^1$  map from  $(r, t) \in D' = \{0 \leq r \leq 1, 0 \leq t \leq \frac{\pi}{2}\}$  to  $(x, y) \in D$ , and by computation

$$\frac{\partial(x, y)}{\partial(r, t)} = \begin{vmatrix} 2r \cos^4 t & -4r^2 \cos^3 t \sin t \\ 2r \sin^4 t & 4r^2 \sin^3 t \cos t \end{vmatrix} = 8r^3 \sin^3 t \cos^3 t.$$

This map is a coordinate transformation from  $\mathring{D}'$  to  $\mathring{D}$ , but the Jacobian vanishes on three edges of the rectangle  $D'$ . However, the substitution

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<sup>3</sup>Strictly speaking, the polar coordinate transform is degenerate at  $r = 0$ , and we should first use a limit argument to bypass the problem caused by this.

formula still works, i.e.

$$\begin{aligned}\iint_D (\sqrt{x} + \sqrt{y}) \, dx dy &= \iint_{D'} r \cdot 8r^3 \sin^3 t \cos^3 t \, dr dt \\ &= \int_0^1 8r^3 dr \int_0^{\frac{\pi}{2}} \sin^3 t \cos^3 t \, dt \\ &= \frac{2}{15}.\end{aligned}\quad (6.3)$$

Strictly speaking, the substitution works since

$$\begin{aligned}\iint_D (\sqrt{x} + \sqrt{y}) \, dx dy &= \lim_{\delta \rightarrow 0^+} \iint_{D_\delta} (\sqrt{x} + \sqrt{y}) \, dx dy \\ &= \iint_{D'_\delta} r \cdot 8r^3 \sin^3 t \cos^3 t \, dr dt \\ &= \iint_{D'} r \cdot 8r^3 \sin^3 t \cos^3 t \, dr dt\end{aligned}\quad (6.4)$$

where  $D'_\delta = \{\delta \leq r \leq 1, \delta \leq t \leq \frac{\pi}{2} - \delta\}$  and  $D_\delta$  is the image under the map  $(r, t) \rightarrow (x, y)$ .

2. *Method 2:* we use the substitution  $(x, y) = (u^2, v^2)$ . Then this map is  $C^1$  from  $D'' = \{u \geq 0, v \geq 0, u + v \leq 1\}$  to  $D$ , and the Jacobian is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 2u & 0 \\ 0 & 2v \end{vmatrix} = 4uv.$$

This map is a coordinate transformation from  $\mathring{D}''$  to  $\mathring{D}$ , but also the Jacobian vanishes on the boundaries  $u = 0$  and  $v = 0$ . By the same reason as the previous case,

$$\iint_D (\sqrt{x} + \sqrt{y}) \, dx dy = \iint_{D''} (u + v) \cdot 4uv \, du dv.$$

**Example 6.6.** Let  $D$  be the region enclosed by the four parabolas  $y^2 = px, y^2 = qx$  and  $x^2 = ay, x^2 = by$  where  $0 < p < q, 0 < a < b$ . Find the double integral  $\iint_D x^2 y^2 \, dx dy$ .

**Solution.** The shape of the region inspires us to consider the substitution  $y^2 = ux, x^2 = vy$ , i.e.  $(x, y) = \left((uv^2)^{\frac{1}{3}}, (u^2v)^{\frac{1}{3}}\right)$  where  $(u, v) \in [p, q] \times [a, b]$ . The Jacobian of this  $C^1$  coordinate transformation is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{1}{3} \left(\frac{v}{u}\right)^{\frac{2}{3}} & \frac{2}{3} \left(\frac{v}{u}\right)^{\frac{1}{3}} \\ \frac{2}{3} \left(\frac{v}{u}\right)^{\frac{1}{3}} & \frac{1}{3} \left(\frac{u}{v}\right)^{\frac{2}{3}} \end{vmatrix} = -\frac{1}{3}.$$

Therefore, we have

$$\begin{aligned}\iint_D x^2 y^2 \, dx dy &= \iint_{u \in [p, q], v \in [a, b]} u^2 v^2 \frac{1}{3} \, du dv \\ &= \frac{1}{27} (q^3 - p^3)(b^3 - a^3).\end{aligned}\quad (6.5)$$

**Example 6.7.** Find the double integral  $\iint_{x^2+y^2 \leq R^2} e^{-x^2-y^2} dx dy$ .

**Solution.** It is difficult to find the antiderivative of the function  $e^{-x^2}$ , so we cannot use the iteration formula directly about  $x$  and  $y$ . Consider the coordinate transformation  $(x, y) = r \cos \theta, y = r \sin \theta$  from  $D' = [0, R] \times [0, 2\pi]$  to  $D = \{x^2 + y^2 \leq R^2\}$ . The Jacobian is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \cos \theta \cdot r \cos \theta - \sin \theta(-r \sin \theta) = r.$$

Even though the Jacobian vanishes at  $r = 0$ , we can use the technique of approximation as before to show that

$$\begin{aligned} \iint_{x^2+y^2 \leq R^2} e^{-x^2-y^2} dx dy &= \iint_{r \in [0, R], \theta \in [0, 2\pi]} e^{-r^2} r dr d\theta \\ &= \int_0^R r e^{-r^2} dr \int_0^{2\pi} d\theta \\ &= \pi(1 - e^{-R^2}). \end{aligned} \quad (6.6)$$

Using the inequality

$$\begin{aligned} \iint_{x^2+y^2 \leq R^2} e^{-x^2-y^2} dx dy &\leq \iint_{|x| \leq R, |y| \leq R} e^{-x^2-y^2} dx dy \\ &= \left( \int_{-R}^R e^{-x^2} dx \right)^2 \leq \iint_{x^2+y^2 \leq 2R^2} e^{-x^2-y^2} dx dy, \end{aligned}$$

we obtain that

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \lim_{R \rightarrow +\infty} \int_{-R}^R e^{-x^2} dx = \sqrt{\pi}.$$

## 7 Triple integral

The theory about Jordan measure and double integral can be easily generalized to 3-dimension and more generally  $n$ -dimension case.

For a measurable set  $V$  in  $\mathbf{R}^3$ , by taking limit of the Riemann sum

$$\sum_{i=1}^n \rho(\xi_i, \eta_i, \zeta_i) \sigma(V_i)$$

as the modulus of the partition  $V = \cup_i V_i$  goes to 0, we can get

$$\iiint_V \rho(x, y, z) d\sigma.$$

The volume element  $d\sigma$  is actually  $d\sigma_3$  which means 3-dimensional Jordan measure, in comparison with the  $d\sigma = d\sigma_2$  in double integral. In practice, 3-dimensional Jordan measure is called volume and therefore  $d\sigma_3$  is usually written as  $dV$ . In terms of Cartesian coordinate  $x, y, z$ , we can write  $dV = dx dy dz$ , and sometimes we simply write  $\int_V f dV$  for triple integral.

**Theorem 7.1.** *Suppose  $f$  is defined on a bounded domain  $V \subset \mathbf{R}^3$ .*

1. *If  $f$  is integrable on  $V$ , then  $f$  is bounded;*
2. *If  $f$  is continuous on  $V$ , then  $f$  is integrable. Furthermore, if  $f$  is bounded on  $V$  and the discontinuities of  $f$  lie on finitely many  $C^1$  surfaces, then  $f$  is integrable.*

We will focus on the computation of triple integrals.

## 7.1 Iterative integration formula

Suppose  $V = I_1 \times I_2 \times I_3 = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$  and we take a rectangular partition:

$$\begin{aligned} T_x : a_1 = x_0 < x_1 < \cdots < x_n = b_1 \\ T_y : a_2 = y_0 < y_1 < \cdots < y_m = b_2 \\ T_z : a_3 = z_0 < z_1 < \cdots < z_k = b_3, \end{aligned}$$

and take the special (aligned) sampling points  $\{(\xi_i, \eta_j, \zeta_l)\}$  with  $1 \leq i \leq n, 1 \leq j \leq m, 1 \leq l \leq k$ . Then, the Riemann sum is

$$S(T, f, \{(\xi_i, \eta_j, \zeta_l)\}) = \sum_{i=1}^n \sum_{j=1}^m \sum_{l=1}^k f(\xi_i, \eta_j, \zeta_l) \Delta x_i \Delta y_j \Delta z_l.$$

On the one hand, we can rearrange the sum as

$$\sum_{i=1}^n \sum_{j=1}^m \left[ \sum_{l=1}^k f(\xi_i, \eta_j, \zeta_l) \Delta z_l \right] \Delta x_i \Delta y_j.$$

By letting  $\|T_z\| \rightarrow 0$ , the interior summation converges to  $\int_{a_3}^{b_3} f(\xi_i, \eta_j, z) dz$ . Then letting  $\|T_x\| + \|T_y\| \rightarrow 0$ , we obtain that the whole summation converges to  $\iint_{[a_1, b_1] \times [a_2, b_2]} dx dy \int_{[a_3, b_3]} f(x, y, z) dz$ .

On the other hand, if we arrange the sum as

$$\sum_{i=1}^n \left[ \sum_{j=1}^m \sum_{l=1}^k f(\xi_i, \eta_j, \zeta_l) \Delta y_j \Delta z_l \right] \Delta x_i,$$

and by letting  $\|T_y\| + \|T_z\| \rightarrow 0$  first, we obtain the limit

$$\sum_{i=1}^n \left[ \iint_{[a_2, b_2] \times [a_3, b_3]} f(\xi_i, y, z) dy dz \right] \Delta x_i,$$

and a further limit  $\|T_x\| \rightarrow 0$  implies the limit is  $\int_{[a_1, b_1]} dx \iint_{[a_2, b_2] \times [a_3, b_3]} f(x, y, z) dy dz$ .

By directly requiring  $\|T_x\|, \|T_y\|, \|T_z\|$  go to 0 simultaneously, the limit is  $\iiint_{I_1 \times I_2 \times I_3} f(x, y, z) dx dy dz$ . Carrying out the argument rigorously as in dimension two, we can prove the following iterative integration formula.

**Theorem 7.2 (Iterative Integration Formula).** Suppose  $f$  is continuous on the bounded domain  $V$ .

- Suppose  $V$  is enclosed by the graphs of two continuous functions  $z = z_1(x, y), (x, y) \in D$  and  $z = z_2(x, y), (x, y) \in D$  where  $z_1(x, y) \leq z_2(x, y)$  and the cylinder over  $\partial D$ , then

$$\iiint_V f(x, y, z) dx dy dz = \iint_D dx dy \int_{z_1(x, y)}^{z_2(x, y)} f(x, y, z) dz.$$

If furthermore,  $D = \{(x, y) | a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\}$  where  $\varphi_1 \leq \varphi_2$  are two continuous function, then

$$\iiint_V f(x, y, z) dx dy dz = \int_a^b dx \int_{\varphi_1(x)}^{\varphi_2(x)} dy \int_{z_1(x, y)}^{z_2(x, y)} f(x, y, z) dz.$$

- Suppose  $V$  projects to an interval  $I$  on  $z$ -axis, and for each  $(0, 0, z)$  the intersection of the plane perpendicular to  $z$ -axis through  $(0, 0, z)$  and  $V$  projects to  $D_z$  on  $xOy$  plane, then

$$\iiint_V f dV = \int_I dz \iint_{D_z} f(x, y, z) dx dy.$$

- If  $V$  can be divided into the union of finitely many regions of the shapes in item 1 and 2, then we can divide the triple integral into the sum of the triple integrals on each of the regions.

**Example 7.3.** Let  $V$  be the tetrahedron enclosed by the four planes  $x = 0, y = 0, z = 0$  and  $x + y + z = 1$ . Find the integral  $\iiint_C \frac{dx dy dz}{(1+x+y+z)^3}$ .

**Solution.** By iterative integration formula,

$$\begin{aligned} \iiint_C \frac{dx dy dz}{(1+x+y+z)^3} &= \iint_{x+y \leq 1} dx dy \int_0^{1-x-y} \frac{dz}{(1+x+y+z)^3} \\ &= \frac{1}{2} \iint_{x \geq 0, y \geq 0, x+y \leq 1} \left( \frac{1}{(1+x+y)^2} - \frac{1}{4} \right) dx dy \\ &= \frac{1}{2} \int_0^1 dx \int_0^{1-x} \left( \frac{1}{(1+x+y)^2} - \frac{1}{4} \right) dy \\ &= \frac{1}{2} \int_0^1 \left( \frac{1}{1+x} - \frac{3-x}{4} \right) dx \\ &= \frac{1}{2} \left( \log 2 - \frac{5}{8} \right). \end{aligned}$$

**Example 7.4.** Let  $V$  be enclosed by the cone  $R^2 z^2 = h^2(x^2 + y^2)$  and  $z = h$ , find the integral  $\iiint_V z dx dy dz$ .

**Example 7.5.** Calculate  $\iiint_{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1} (x^2 + y^2 + z^2) dx dy dz$ .

**Solution.** For the term  $x^2$ , we can try  $\int_{\frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1} dy dz \int_{-a\sqrt{1-\frac{y^2}{b^2}-\frac{z^2}{c^2}}}^{a\sqrt{1-\frac{y^2}{b^2}-\frac{z^2}{c^2}}} x^2 dx$  or  $\int_{-a}^a dx \int_{\frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1-\frac{x^2}{a^2}} x^2 dy dz$ . (Compare the difficulty in calculations of these two methods).

For other terms  $y^2$  and  $z^2$  we can treat similarly.

**Example 7.6.** Let  $V$  be enclosed by the plane  $z = h$  and the paraboloid  $x^2 + y^2 \leq 4z$ , find  $\iiint_V \frac{dx dy dz}{1+x^2+y^2}$ .

**Solution.**

$$\begin{aligned} \int_0^h dz \iint_{x^2+y^2 \leq 4z} \frac{dx dy}{1+x^2+y^2} &= \int_0^h dz \int_0^{2\sqrt{z}} \int_0^{2\pi} \frac{r dr d\theta}{1+r^2} \\ &= \pi \int_0^h \log(1+4z) dz \\ &= \pi \left[ \left( h + \frac{1}{4} \right) \log(1+4h) - h \right] \end{aligned}$$

## 7.2 Substitution Formula

Now we consider the coordinate transformation in  $\mathbf{R}^3$ , suppose  $x = x(u, v, w)$ ,  $y = y(u, v, w)$ ,  $z = z(u, v, w)$  maps the open domain  $V'$  to the open domain  $V$  in one to one way, and suppose  $\partial V, \partial V'$  both consists of finitely many  $C^1$  surface. Suppose  $\frac{\partial(x, y, z)}{\partial(u, v, w)} \neq 0$  on  $V'$ .

A tiny rectangular region  $\Delta u, \Delta v, \Delta w$  is mapped to a curved region  $\Delta x, \Delta y, \Delta z$  in  $V$ . The volume of the latter is approximately  $|\mathbf{r}'_u \Delta u \cdot (\mathbf{r}'_v \Delta v \times \mathbf{r}'_w \Delta w)| = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| \Delta u \Delta v \Delta w$ . The substitution formula for triple integral would read as

$$\begin{aligned} \iiint_V f(x, y, z) dx dy dz \\ = \iiint_{V'} f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw. \end{aligned} \quad (7.1)$$

A particular useful substitution is spherical coordinate transformation  $x = r \sin \theta \cos \varphi$ ,  $y = r \sin \theta \sin \varphi$ ,  $z = r \cos \theta$ . The Jacobian is

$$\begin{vmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{vmatrix} = r^2 \sin \theta \cos^2 \theta + r^2 \sin^3 \theta = r^2 \sin \theta.$$

**Example 7.7.** Calculate the triple integral  $\iiint_V (x^2 + y^2 + z^2) dx dy dz$  where  $V$  is enclosed by the cone  $z = \sqrt{x^2 + y^2}$  and  $x^2 + y^2 + z^2 = R^2$ .

**Solution.** It is easy to recognize that  $V$  is represented by the region  $V' : 0 \leq r \leq R, 0 \leq \theta \leq \frac{\pi}{4}, 0 \leq \varphi \leq 2\pi$ . By substitution, we have

$$\begin{aligned} \iiint_V (x^2 + y^2 + z^2) dx dy dz &= \iiint_{r \in [0, R], \theta \in [0, \frac{\pi}{4}], \varphi \in [0, 2\pi]} r^2 \cdot r^2 \sin \theta dr d\theta d\varphi \\ &= \frac{2\pi}{5} R^5 \left(1 - \frac{\sqrt{2}}{2}\right). \end{aligned}$$

**Example 7.8.** Find the volume of the domain enclosed by the surface  $(x^2 + y^2 + z^2)^2 = a^3 z$  ( $a > 0$ ).

**Solution.** The surface is rotational symmetric along the  $z$ -axis, with the generating curve  $(y^2 + z^2)^2 = a^3 z$  in the  $yOz$  plane.

Around  $(0, 0)$ , we can solve that

$$y = \left(a^{\frac{3}{2}} z^{\frac{1}{2}} - z^2\right)^{\frac{1}{2}} = a^{\frac{3}{4}} z^{\frac{1}{4}} \left(1 - \frac{1}{2} a^{-\frac{3}{2}} z^{\frac{3}{2}} + O(z^3)\right),$$

which means that  $z = \frac{1}{a^3} y^4 + O(y^6)$ . Similarly, around  $(0, a)$  we can solve that

$$\begin{aligned} y^2 &= \left(a^3(z - a) + a^4\right)^{\frac{1}{2}} - z^2 \\ &= \left(a^3(z - a) + a^4\right)^{\frac{1}{2}} - a^2 - 2a(z - a) - (z - a)^2 \\ &= \left(\frac{a^3}{(a^3(z - a) + a^4)^{\frac{1}{2}} + a^2} - 2a\right)(z - a) - (z - a)^2 \end{aligned}$$

which means that  $z = a - \frac{2}{3a} y^2 + O(y^4)$ . This region is of some “egg” type.

If we use spherical coordinate, the equation of the surface is given by  $r^3 = a^3 \cos \theta$  with  $\theta \in [0, \frac{\pi}{2}], \varphi \in [0, 2\pi]$ . Thus, the volume of  $V$  is

$$\begin{aligned} \iiint_V dx dy dz &= \iiint_{\theta \in [0, \frac{\pi}{2}], \varphi \in [0, 2\pi], 0 \leq r \leq a^{\frac{1}{3}} \sqrt[3]{\cos \theta}} r^2 \sin \theta dr d\theta d\varphi \\ &= \int_0^{\frac{\pi}{2}} \sin \theta d\theta \int_0^{\sqrt[3]{a \cos \theta}} r^2 dr \int_0^{2\pi} d\varphi \\ &= \frac{\pi a^3}{3}. \end{aligned}$$

**Example 7.9.** Find the volume of the region enclosed by the surface  $\left(\frac{x}{a}\right)^{\frac{2}{3}} + \left(\frac{y}{b}\right)^{\frac{2}{3}} + \left(\frac{z}{c}\right)^{\frac{2}{3}} = 1$ .

**Solution.** Using  $x = au^3, y = bv^3, z = cw^3$ , then the Jacobian is  $\frac{\partial(x, y, z)}{\partial(u, v, w)} = 27abcu^2v^2w^2$ . Even though it vanishes on the union of the three coordinate

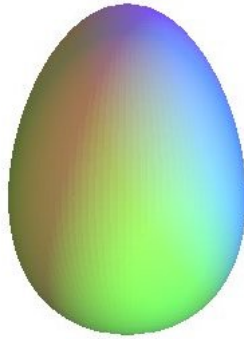


Figure 2: Egg surface

planes, but it is easily seen that the substitution still works

$$\begin{aligned}
 \iiint_V dx dy dz &= 8 \iiint_{u^2+v^2+w^2 \leq 1, u,v,w \geq 0} 27abc u^2 v^2 w^2 du dv dw \\
 &= 27abc \iiint_{u^2+v^2+w^2 \leq 1} u^2 v^2 w^2 du dv dw \\
 &= 27abc \iiint_{r \in [0,1], \theta \in [0,\pi], \varphi \in [0,2\pi]} r^6 \sin^4 \theta \cos^2 \theta \cos^2 \varphi \sin^2 \varphi \cdot r^2 \sin \theta dr d\theta d\varphi \\
 &= \frac{4}{35} \pi abc.
 \end{aligned}$$

### 7.3 \*Some examples from Physics\*

We use the pair  $(V, \rho)$  where  $V$  is a domain in  $\mathbf{R}^3$  and  $\rho : V \rightarrow \mathbf{R}_{\geq 0}$  is a nonnegative function on  $V$  to represent a rigid body occupying the space region  $V$  with density function  $\rho$ .

- 1) [\[Center of Gravity\]](#)

$$\mathbf{r}_G = \frac{\iiint_V \mathbf{r} \rho(\mathbf{r}) dV}{\iiint_V \rho(\mathbf{r}) dV},$$

where the integral of a vector-valued function is understood in terms of its components, i.e.  $\mathbf{r}_G = (x_G, y_G, z_G)$  where

$$x_G = \frac{\iiint_V x \rho(\mathbf{r}) dV}{\iiint_V \rho(\mathbf{r}) dV}, \quad y_G = \frac{\iiint_V y \rho(\mathbf{r}) dV}{\iiint_V \rho(\mathbf{r}) dV}, \quad z_G = \frac{\iiint_V z \rho(\mathbf{r}) dV}{\iiint_V \rho(\mathbf{r}) dV}.$$

- 2) [\[Moment of inertia of rigid body\]](#) The moment of inertia of  $(V, \rho)$  relative



to the point  $\mathbf{r}_0$  is

$$\begin{aligned}\iiint_V |\mathbf{r} - \mathbf{r}_0|^2 \rho(\mathbf{r}) dV &= \iiint_V \left[ |\mathbf{r} - \mathbf{r}_G|^2 + |\mathbf{r}_0 - \mathbf{r}_G|^2 \right. \\ &\quad \left. - 2(\mathbf{r} - \mathbf{r}_G) \cdot (\mathbf{r}_0 - \mathbf{r}_G) \right] \rho(\mathbf{r}) dV \\ &= I_G + |\mathbf{r}_0 - \mathbf{r}_G|^2 M.\end{aligned}$$

This means that the rigid body has minimum moment of inertia with respect to *the center of gravity*. In other words, this variational characterization gives another definition of the *center of gravity*.

Similar to this, we can define the moment of inertia with respect to some rotation axis  $\ell$  in  $\mathbf{R}^3$ , for instance, the moment of inertia with respect to  $x$ -axis is

$$I_x = \iiint_V (y^2 + z^2) \rho(x, y, z) dx dy dz.$$

- 3) [[Gravity of a body](#)] We consider the gravity of the body  $(V, \rho)$  acting on a point mass  $m$  at  $\mathbf{r}_0 \in \mathbf{R}^3$ :

$$\mathbf{F} = \iiint_V \frac{Gm}{|\mathbf{r} - \mathbf{r}_0|^3} (\mathbf{r} - \mathbf{r}_0) \rho(\mathbf{r}) dV.$$

More general, if we have two bodies  $(V_1, \rho_1)$  and  $(V_2, \rho_2)$ , then the gravity put on the body  $(V_2, \rho_2)$  by the body  $(V_1, \rho_1)$  is given by the iterative triple integral:

$$\mathbf{F} = G \iiint_{V_2} \rho_2(\mathbf{r}_2) dV(\mathbf{r}_2) \iiint_{V_1} \frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|^3} \rho_1(\mathbf{r}_1) dV(\mathbf{r}_1).$$

## 7.4 Integration of $n$ -variables

We can easily generalize the double integral and triple integral to the notion of  $n$ -dimensional integral. For instance, the above gravity can also be understood as

$$\mathbf{F} = G \int_{V_1 \times V_2} \frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|^3} \rho_1(\mathbf{r}_1) \rho_2(\mathbf{r}_2) dV(\mathbf{r}_1, \mathbf{r}_2).$$

We can do iterative integration formula to

$$\int_V f dV = \int_D dx_1 \cdots dx_{n-1} \int_{\varphi_1(x_1, \dots, x_{n-1})}^{\varphi(x_1, \dots, x_n)} f(x_1, \dots, x_n) dx_n. \quad (7.2)$$

$$\begin{aligned}\int \cdots \int_V f(x_1, \dots, x_n) dx_1 \cdots dx_n \\ = \int \cdots \int_{V'} f(x_1(u_1, \dots, u_n), \dots, x_n(u_1, \dots, u_n)) \left| \frac{\partial(x_1, \dots, x_n)}{\partial(u_1, \dots, u_n)} \right| du_1 \cdots du_n.\end{aligned} \quad (7.3)$$

**Example 7.10 (Volume of  $n$ -simplex).** We call the set

$$\Delta_n(a) = \{(x_1, \dots, x_n) | x_1, \dots, x_n \geq 0, x_1 + \dots + x_n \leq a\}$$

the  $n$ -simplex. The volume is

$$\begin{aligned} \sigma(\Delta_n(a)) &= \int_{\Delta_n(a)} d\sigma \\ &=_{x_i=at_i} \int_{\Delta_n(1)} a^n d\sigma \\ &= a^n \int_0^1 dt_n \int_{t_1, \dots, t_{n-1} \geq 0, t_1 + \dots + t_{n-1} \leq 1-t_n} dt_1 \cdots dt_{n-1} \\ &= a^n \int_0^1 (1-t)^{n-1} \sigma(\Delta_{n-1}(1)) dt \\ &= \frac{a^n}{n} \sigma(\Delta_{n-1}) \\ &= \frac{a^n}{n!} \end{aligned} \tag{7.4}$$

**Example 7.11 (Volume of  $n$ -ball).** Consider the  $n$ -ball of radius  $a$ :

$$B_n(a) = \{(x_1, \dots, x_n) | x_1^2 + \dots + x_n^2 \leq a^2\}.$$

The volume is

$$\sigma(B_n(a)) = \int_{B_n(a)} d\sigma = a^n \int_{B_n(1)} d\sigma = a^n \sigma(B_n(1)). \tag{7.5}$$

By projecting the  $n$ -ball to the  $x_n$ -axis, the iterative integration formula reads

$$\begin{aligned} \sigma(B_n(1)) &= \int_{-1}^1 dx_n \int_{x_1^2 + \dots + x_{n-1}^2 \leq 1-x_n^2} dx_1 \cdots dx_{n-1} \\ &= \int_{-1}^1 (1-x_n^2)^{\frac{n-1}{2}} dx_n \cdot \sigma(B_{n-1}(1)) \\ &= 2\sigma(B_{n-1}(1)) \int_0^1 \cos^n \theta d\theta. \end{aligned}$$

If we project the  $n$ -ball to the two-plane spanned by the  $x_{n-1}$  and  $x_n$ -axis, then we get

$$\begin{aligned} \sigma(B_n(1)) &= \int_{u^2+v^2 \leq 1} dudv \int_{x_1^2 + \dots + x_{n-2}^2 \leq 1-u^2-v^2} dt_1 \cdots dt_{n-2} \\ &= \int_{u^2+v^2 \leq 1} (1-u^2-v^2)^{\frac{n-2}{2}} \sigma(B_{n-2}(1)) dudv \\ &= \sigma(B_{n-2}(1)) \int_0^1 \int_0^{2\pi} (1-r^2)^{\frac{n-2}{2}} r dr d\theta \\ &= \frac{2\pi}{n} \sigma(B_{n-2}(1)). \end{aligned}$$

Using the fact that  $\sigma(B_1(1)) = 2$  and  $\sigma(B_2(1)) = \pi$ , we can find the volume of all  $n$ -ball.

The volume of the  $n$ -sphere is left to next chapter about surface integral.